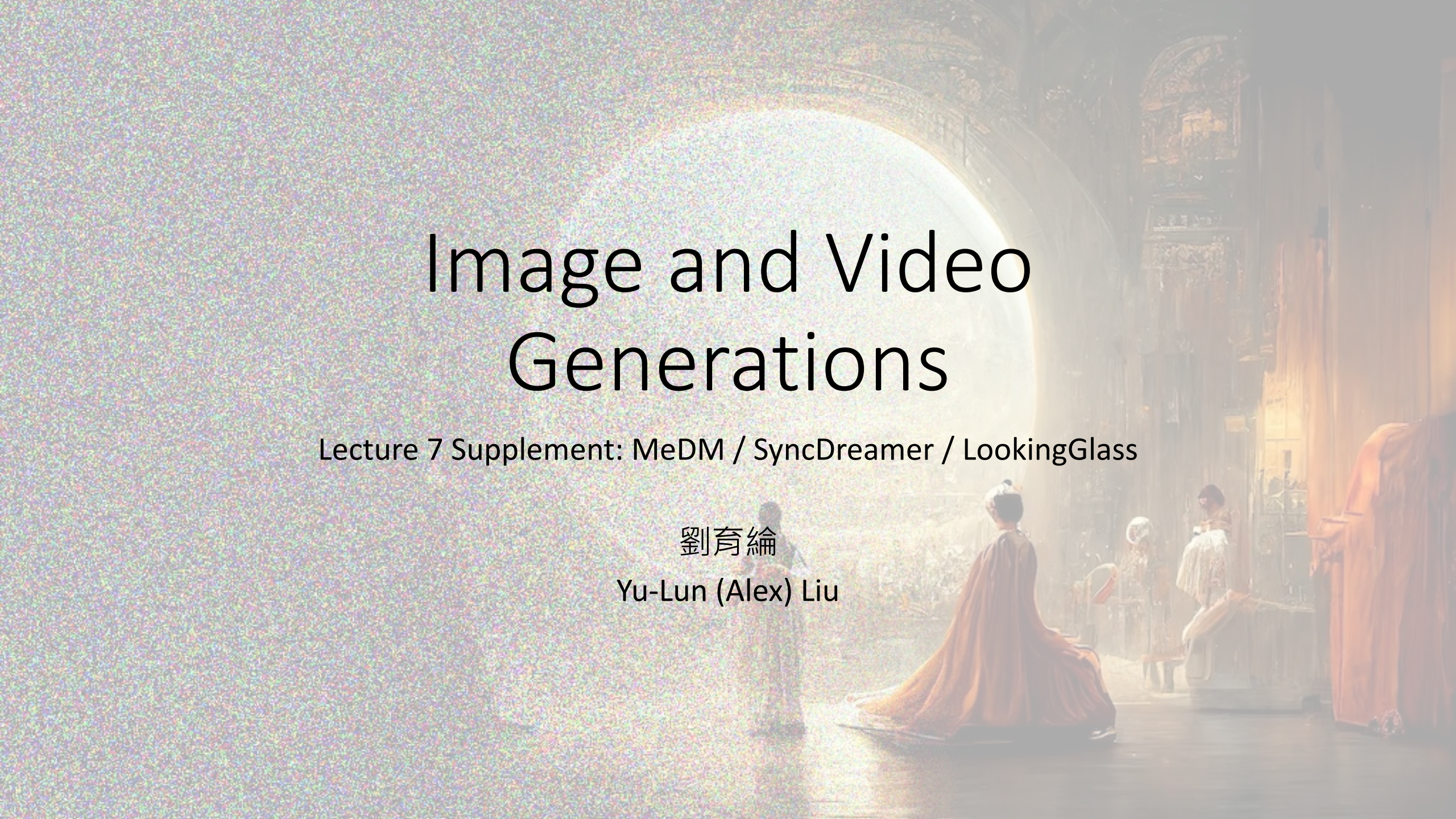


Image and Video Generations

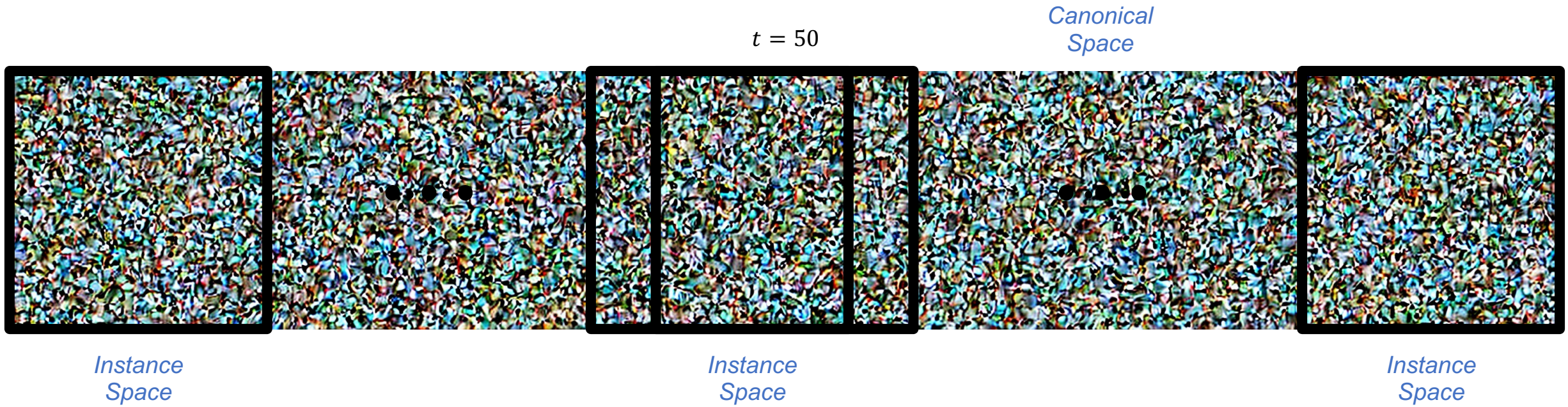
Lecture 7 Supplement: MeDM / SyncDreamer / LookingGlass

劉育綸

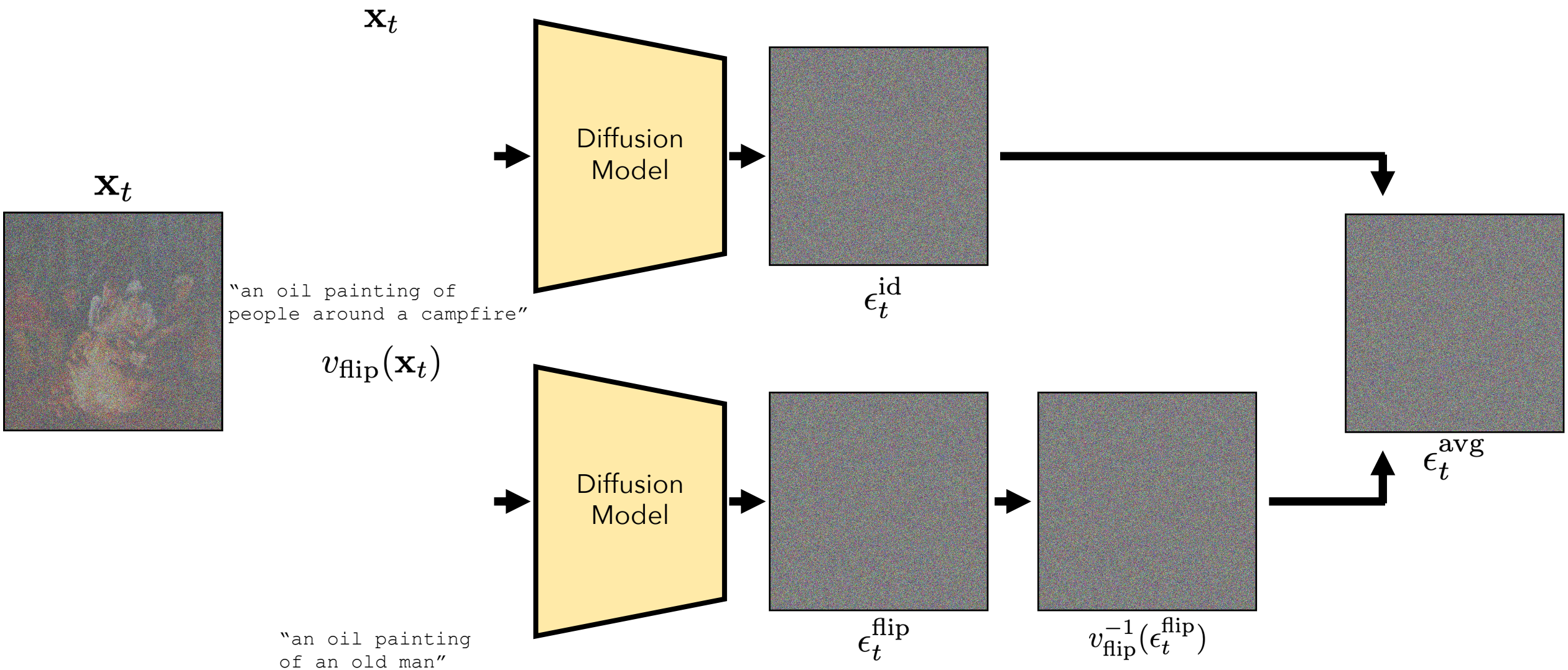
Yu-Lun (Alex) Liu

Recap: MultiDiffusion

- Multiple instances are defined on the canonical space with overlaps.
- Projection is defined as cropping.

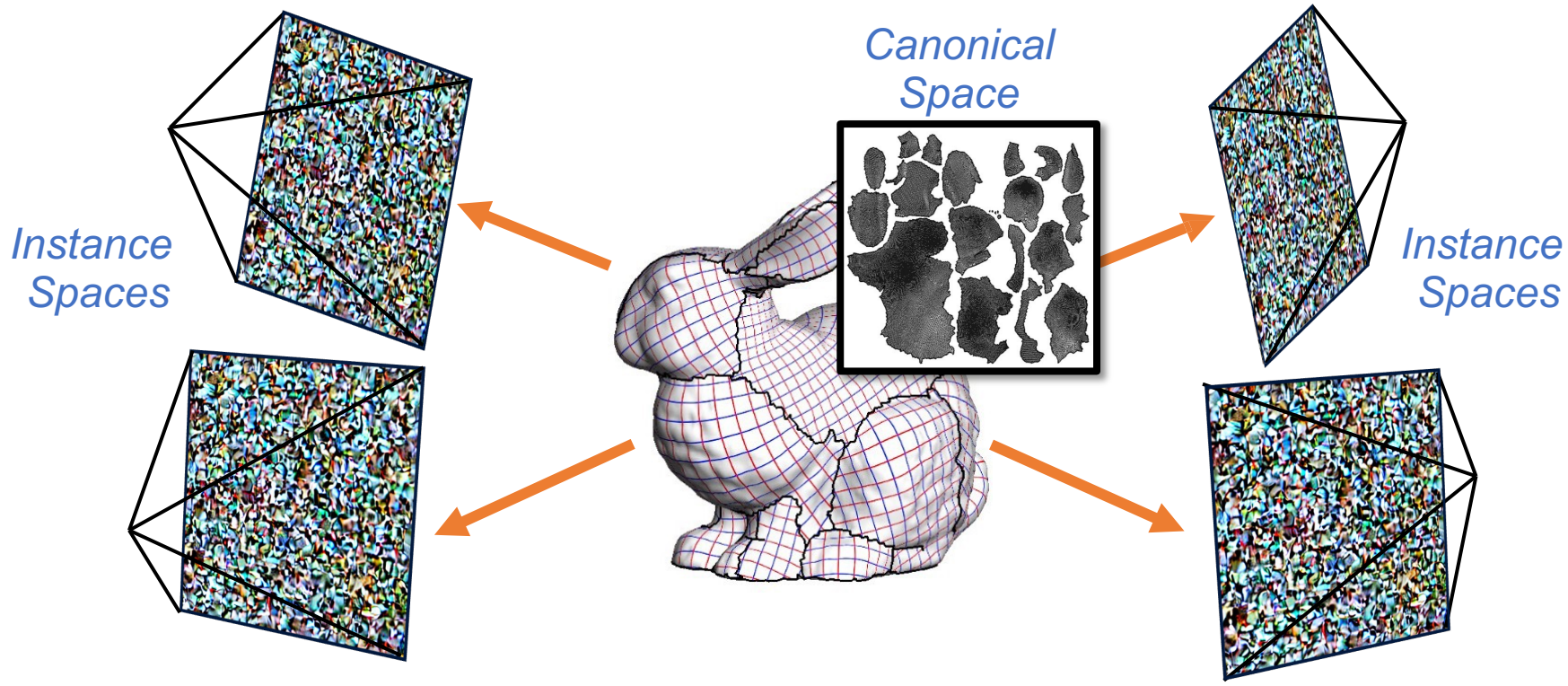


Recap: Visual Anagrams



Recap: Mesh Textures

Projection: Mapping canonical -> instance for each view.



Recap: Pros and Cons

(+) Synchronization generates better quality outputs compared to SDS when strong conditions are given.

(−) Synchronization fails to converge when conditions are weak.

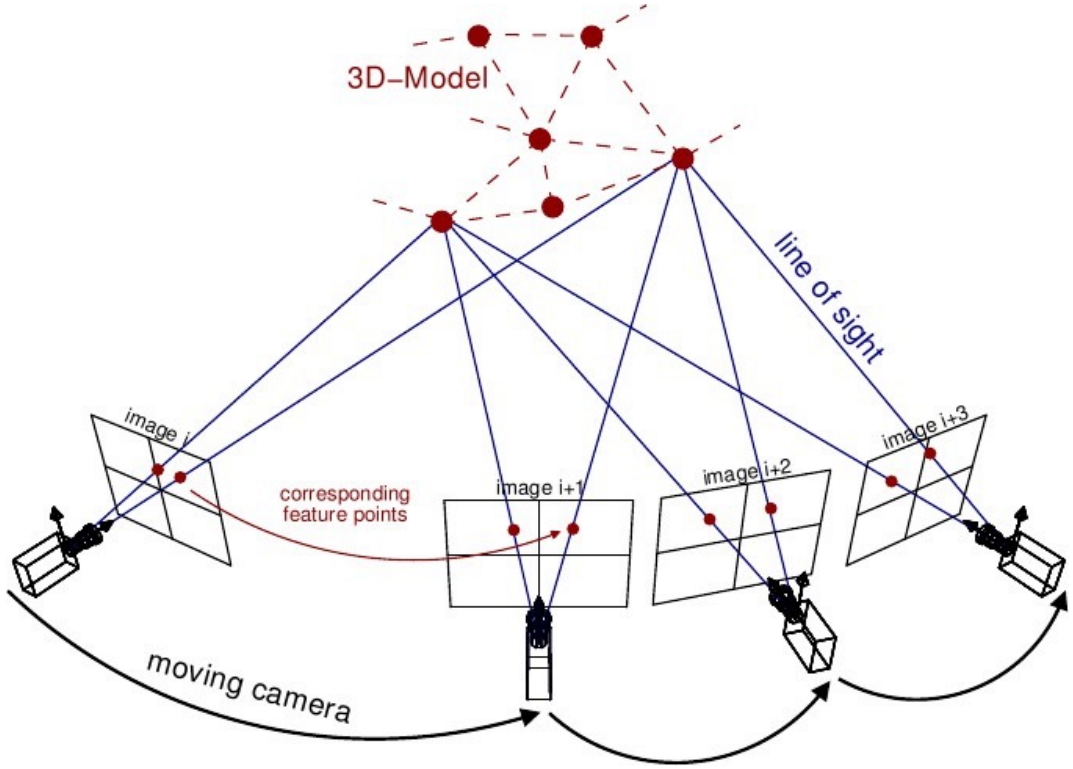
(−) Synchronization also fails to converge for 3D generation.

所以很難使用在 3D generation 這種需要連幾何形狀一起生成，condition 較少的任務

Recap: MVDiffusion

Assumptions

- The mapping between instance and canonical spaces is specified.
- The correspondence across instance images is also defined.



Recap: Limitations: 360° Panoramas

The mappings of instance images to the canonical space are not square but distorted squares.



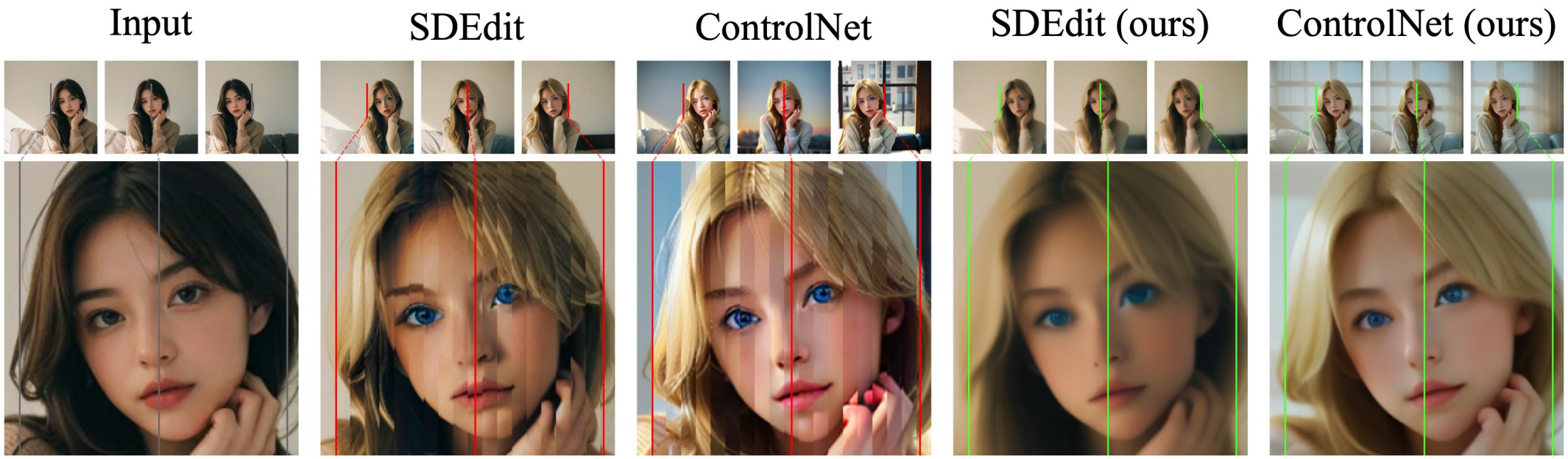
MeDM

Chu, Ernie, et al. "MeDM: Mediating Image Diffusion Models for Video-to-Video Translation with Temporal Correspondence Guidance." *AAAI*. 2024.

Video-to-video Translation

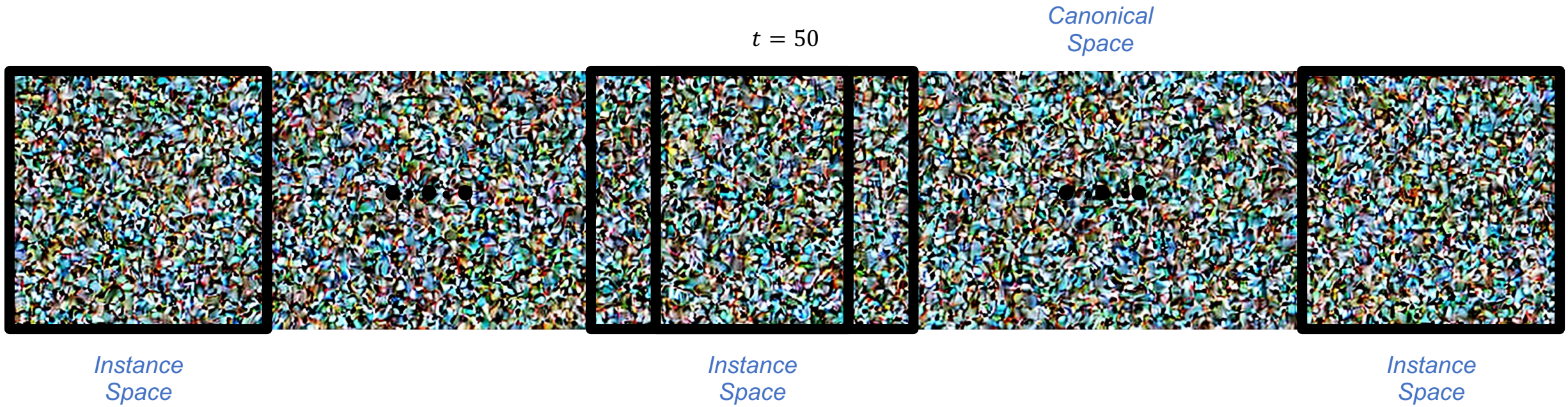


Video-to-video Translation using an **Image Diffusion Model**



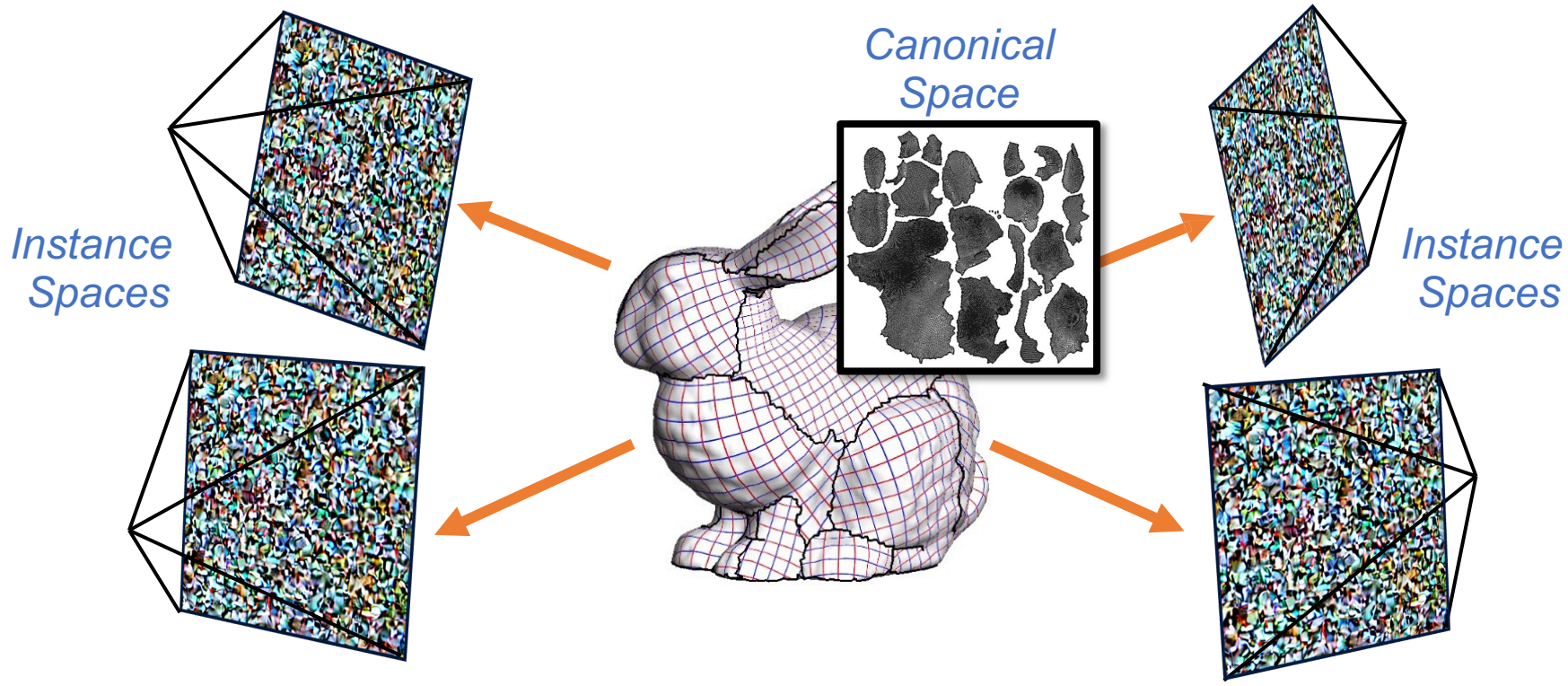
Recap: MultiDiffusion

- Multiple instances are defined on the canonical space with overlaps.
- Projection is defined as cropping.

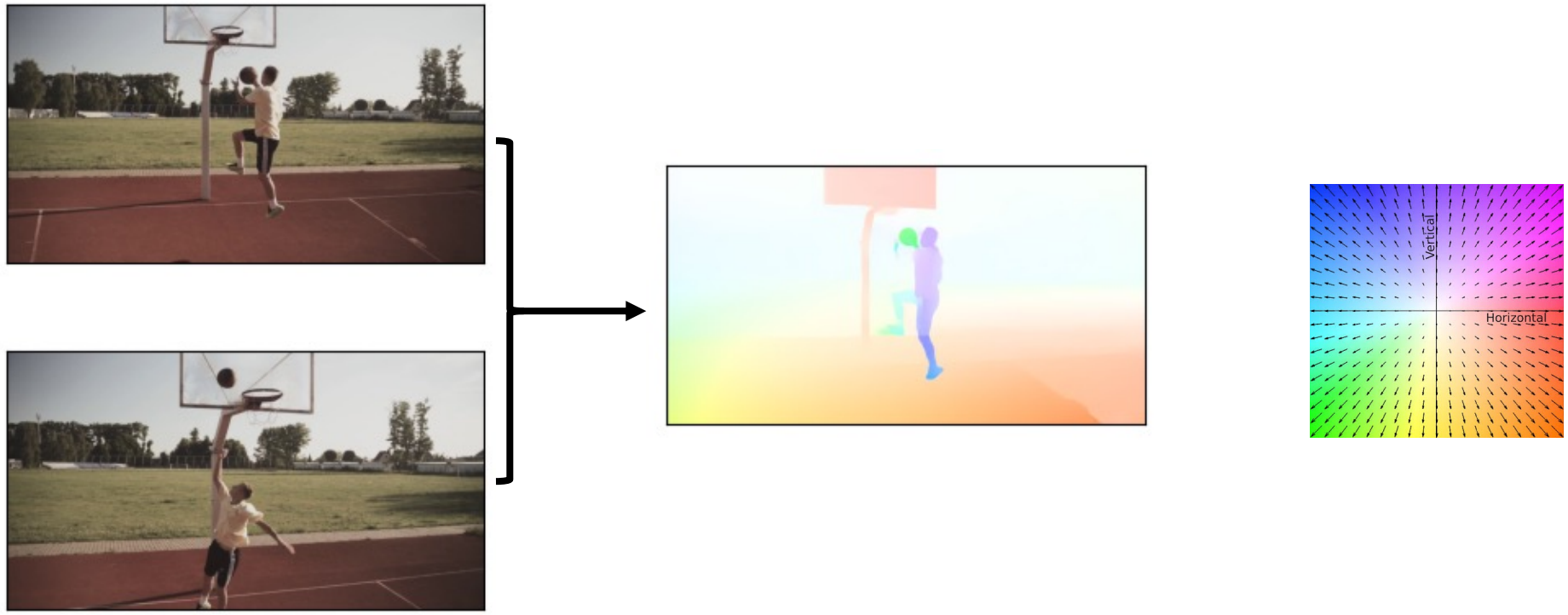


Recap: Mesh Textures

Projection: Mapping canonical -> instance for each view.

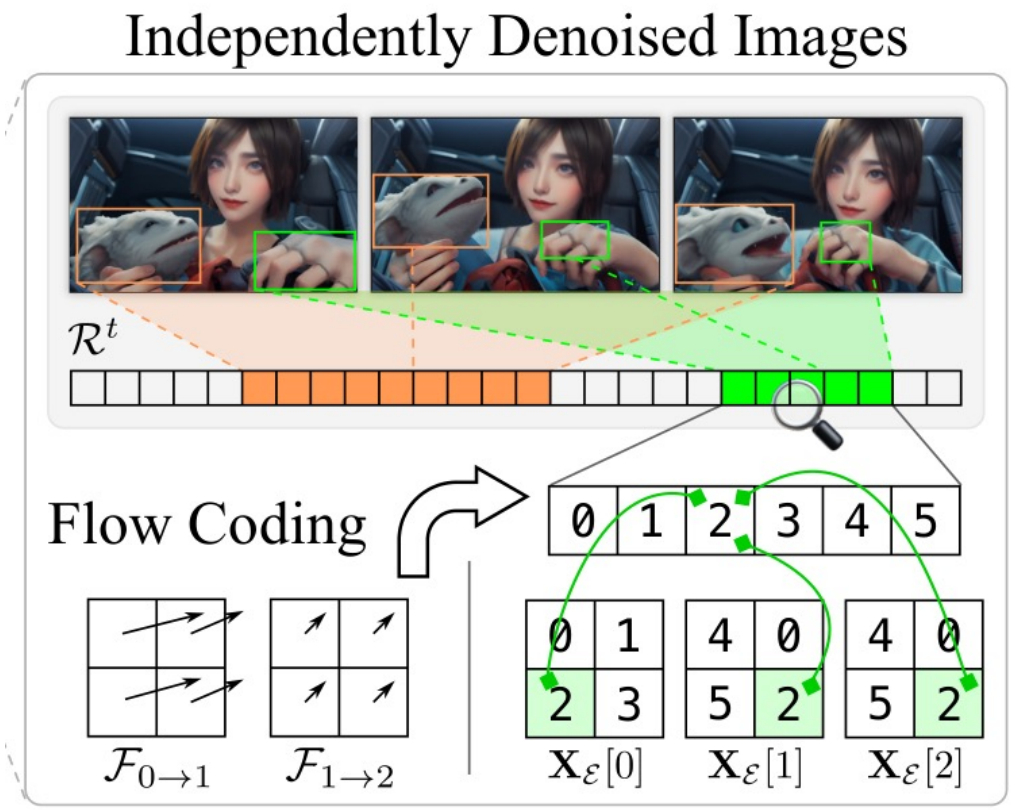


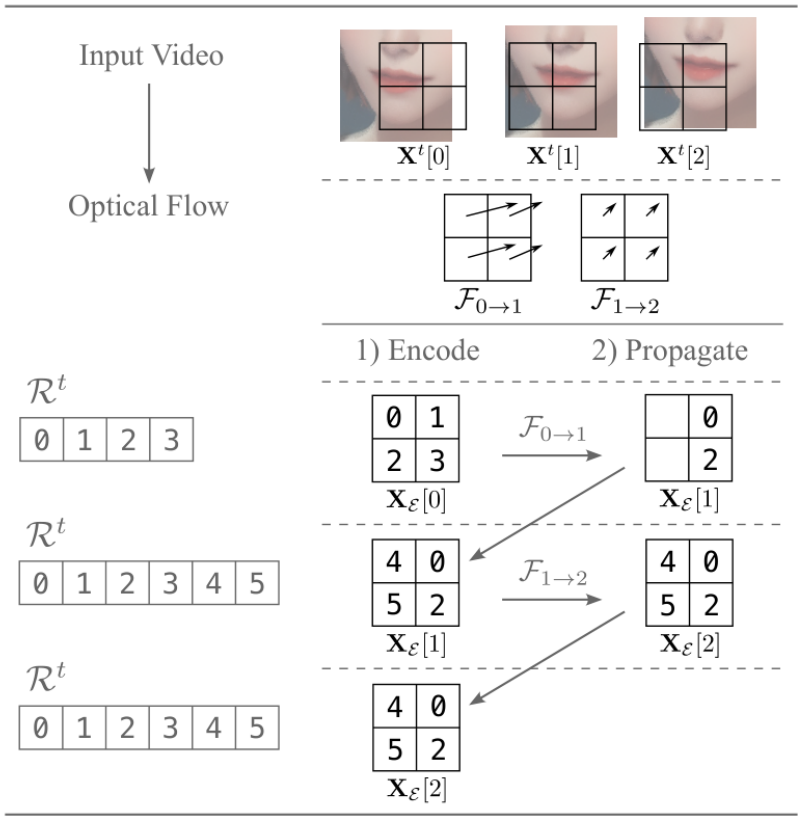
Video Correspondence: Optical Flow



Teed, Zachary, and Jia Deng. "RAFT: Recurrent All-Pairs Field Transforms for Optical Flow." *ECCV.* 2020.

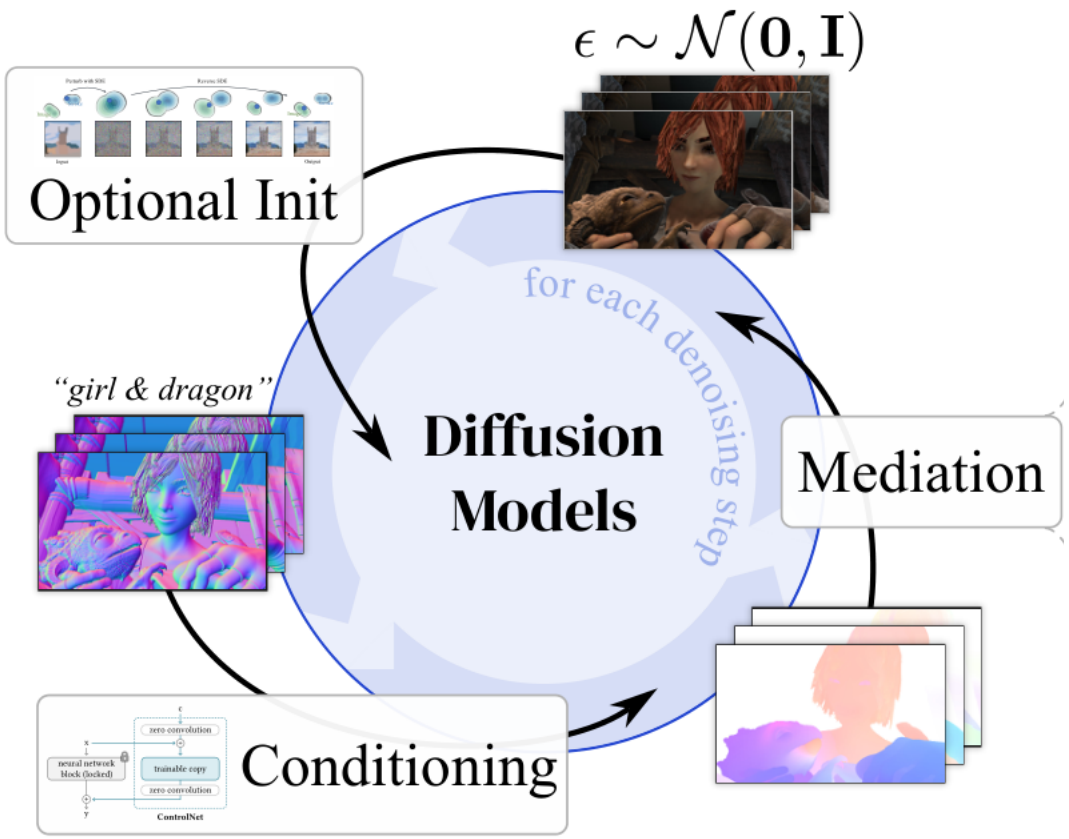
Method: Flow Coding





Unproject: 依 index 寫回 R，Project: 依 index 從 R 取值

Pixel Space MultiDiffusion

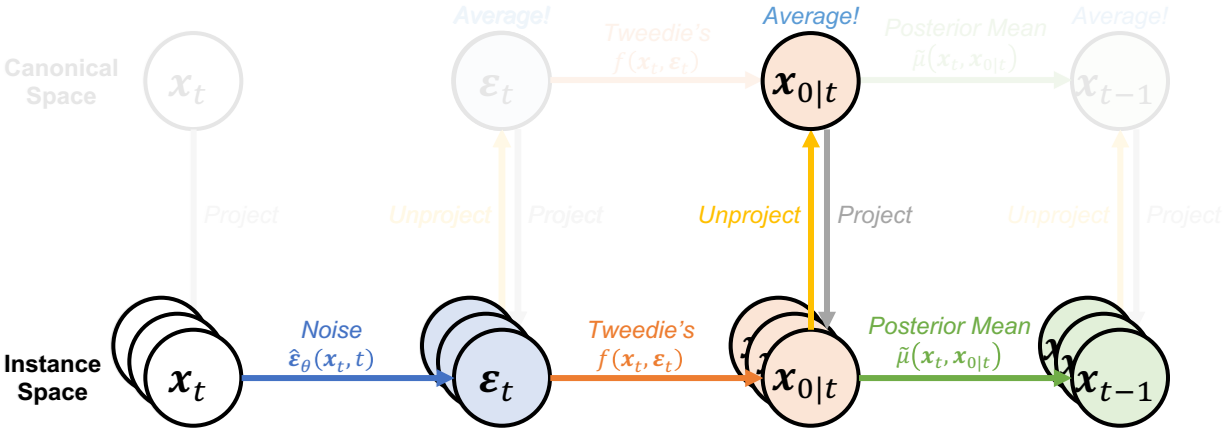


For each denoising step t:

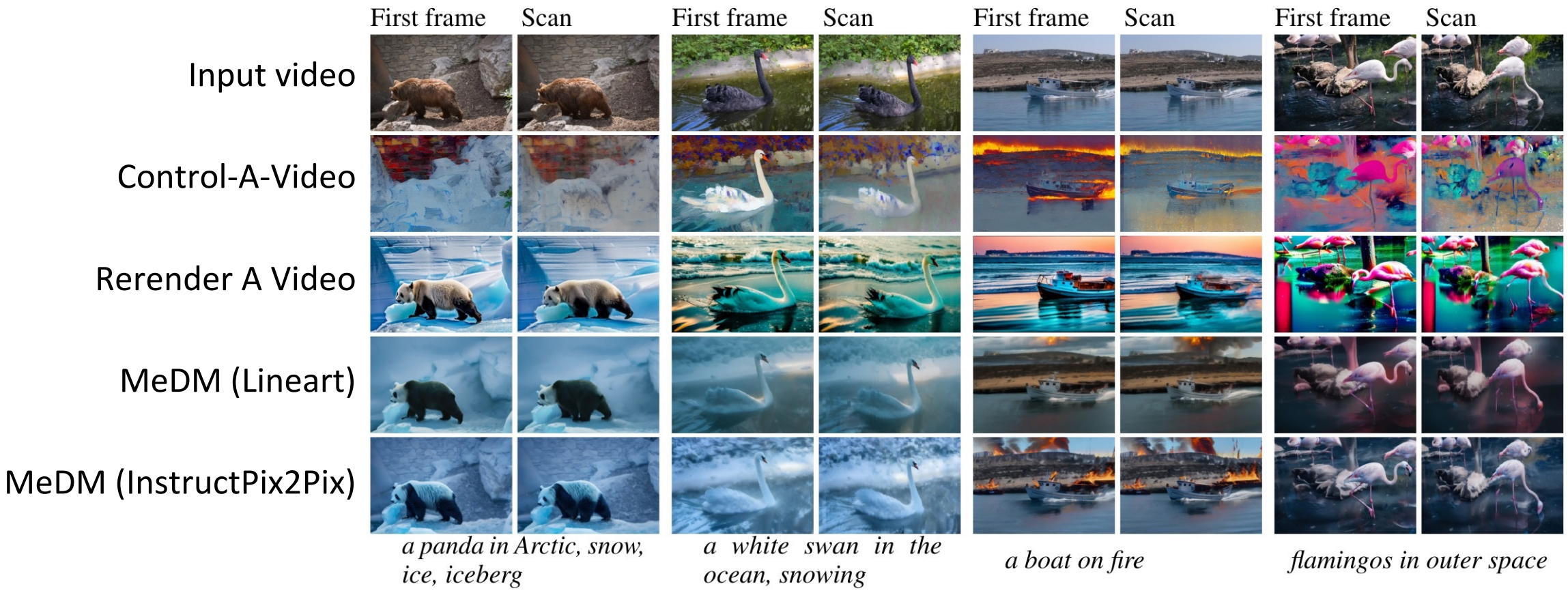
1. 用 ControlNet 或 SDEdit 來 predict noise
2. Tweedie's formula 外推出乾淨的 $\hat{\boldsymbol{x}}_0$，再用 VAE decoder decode 回 pixel
3. 同一個 flow coding index 的 pixel 計算平均 pixel RGB
4. Encode 回 latent，add noise 回到 $\boldsymbol{x}_{t-1}$

Possible Approaches

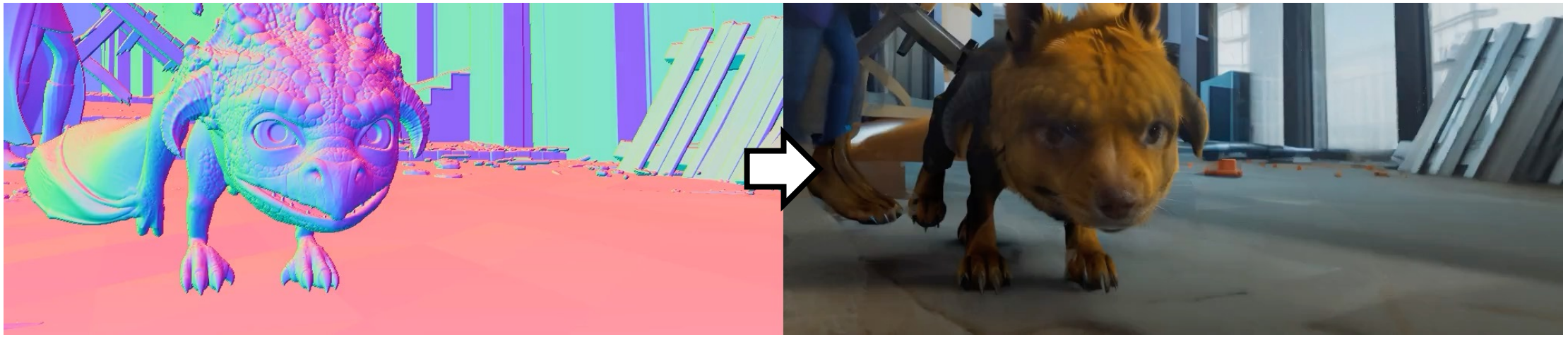
1. Instance space denoising
    1. (Case 1) $\boldsymbol{\varepsilon}_t$ averaging
    2. (Case 2) $\boldsymbol{x}_{0|t}$ averaging
    3. (Case 3) $\boldsymbol{x}_t$ averaging
2. Canonical space denoising
    1. (Case 4) $\boldsymbol{\varepsilon}_t$ averaging
    2. (Case 5) $\boldsymbol{x}_{0|t}$ averaging
    3. (Case 6) $\boldsymbol{x}_t$ averaging



Experiments



Results: Video Rendering



Results: Video-to-video Translation (SDEdit)



Results: Text-Guided Video Edit



Results: Video Anonymization



Summary

只有 image diffusion model，不做任何 training/finetuning，做到 temporally consistent 的 video-to-video translation

利用 optical flow 來建立 video **correspondence**，而且用 flow coding 來建立 project/unproject，應用 MultiDiffusion，使 image diffusion model 也可以用在 video 上做到 editing

SyncDreamer

Liu, Yuan, et al. "SyncDreamer: Generating Multiview-consistent Images from a Single-view Image." *ICLR*. 2024.

Single Image -> Multiview-consistent Images

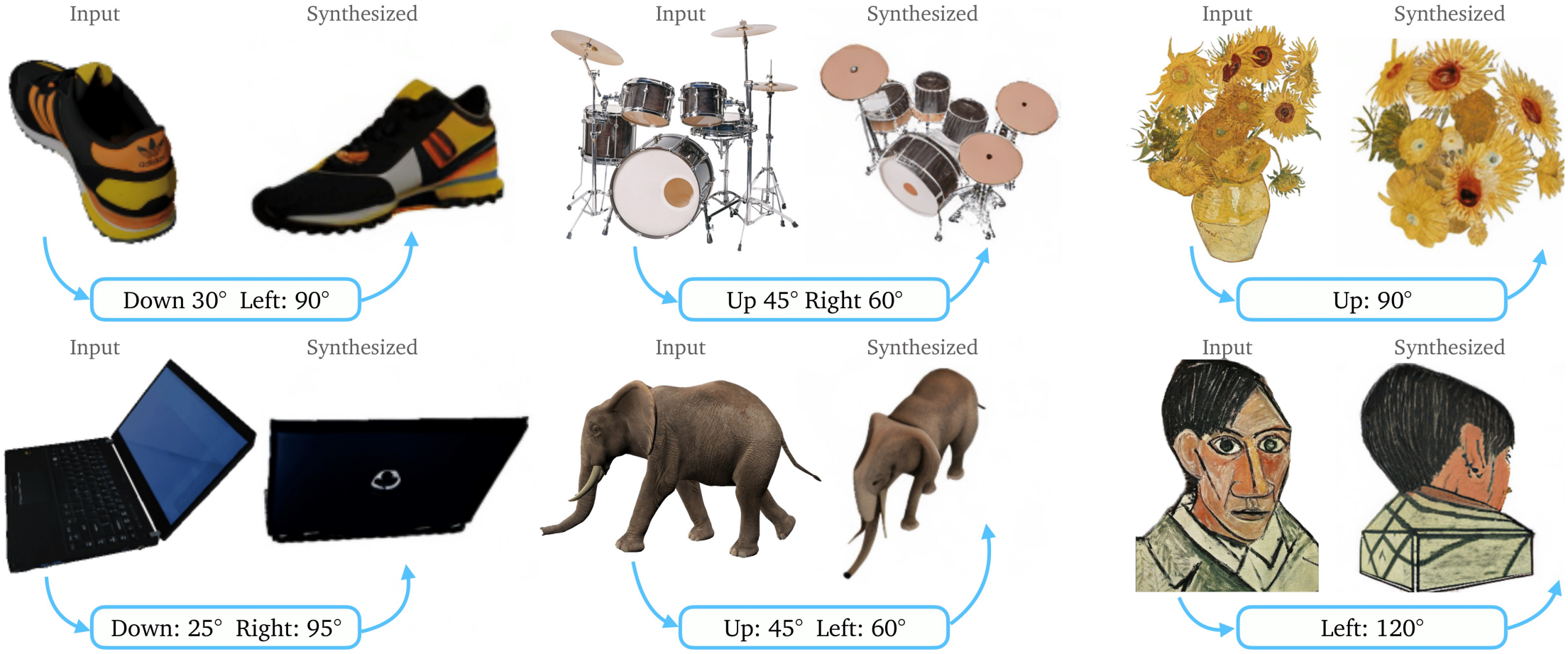


Input

Generated multiview-consistent images

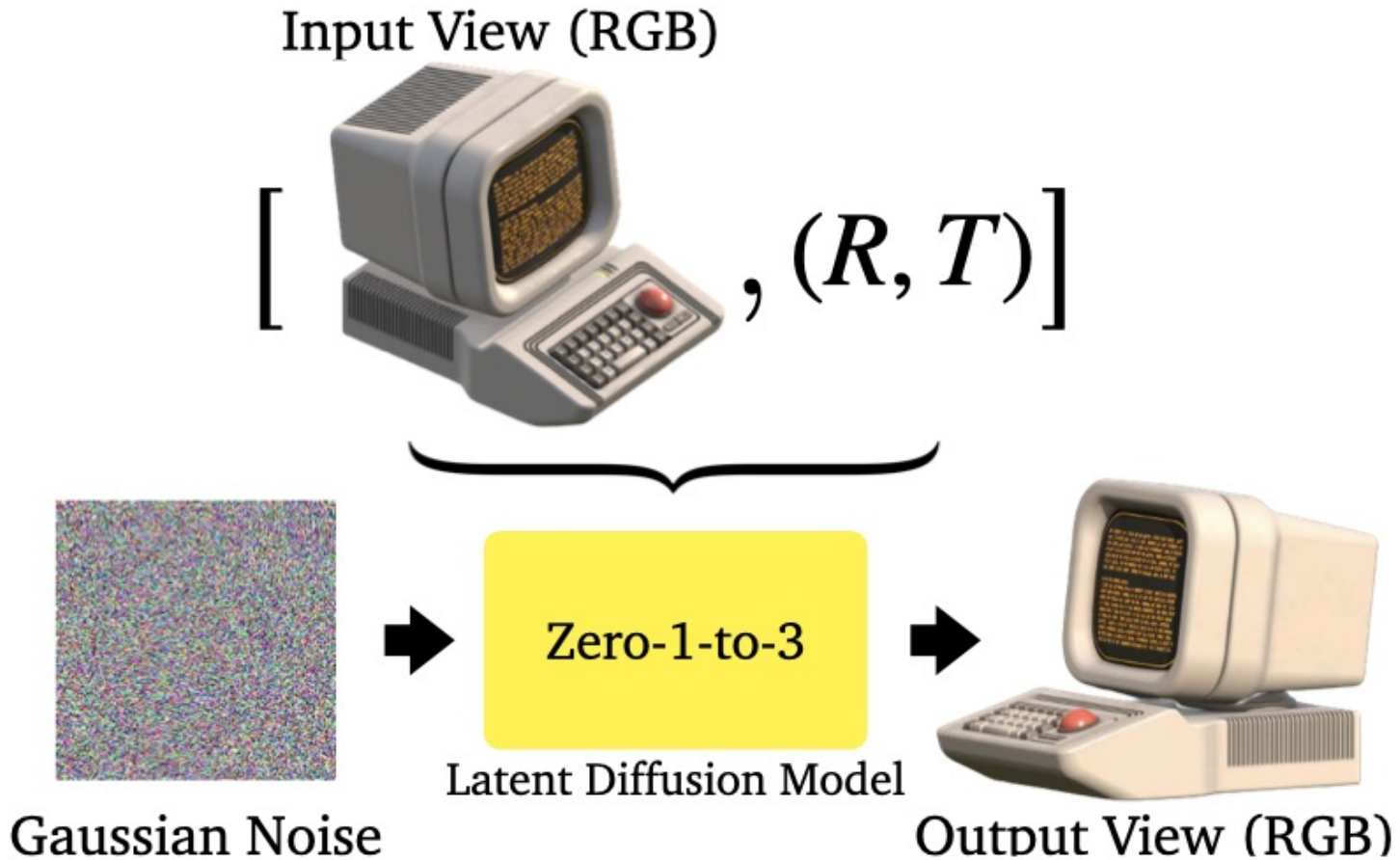
Rendering & Mesh

Preliminary: Zero123



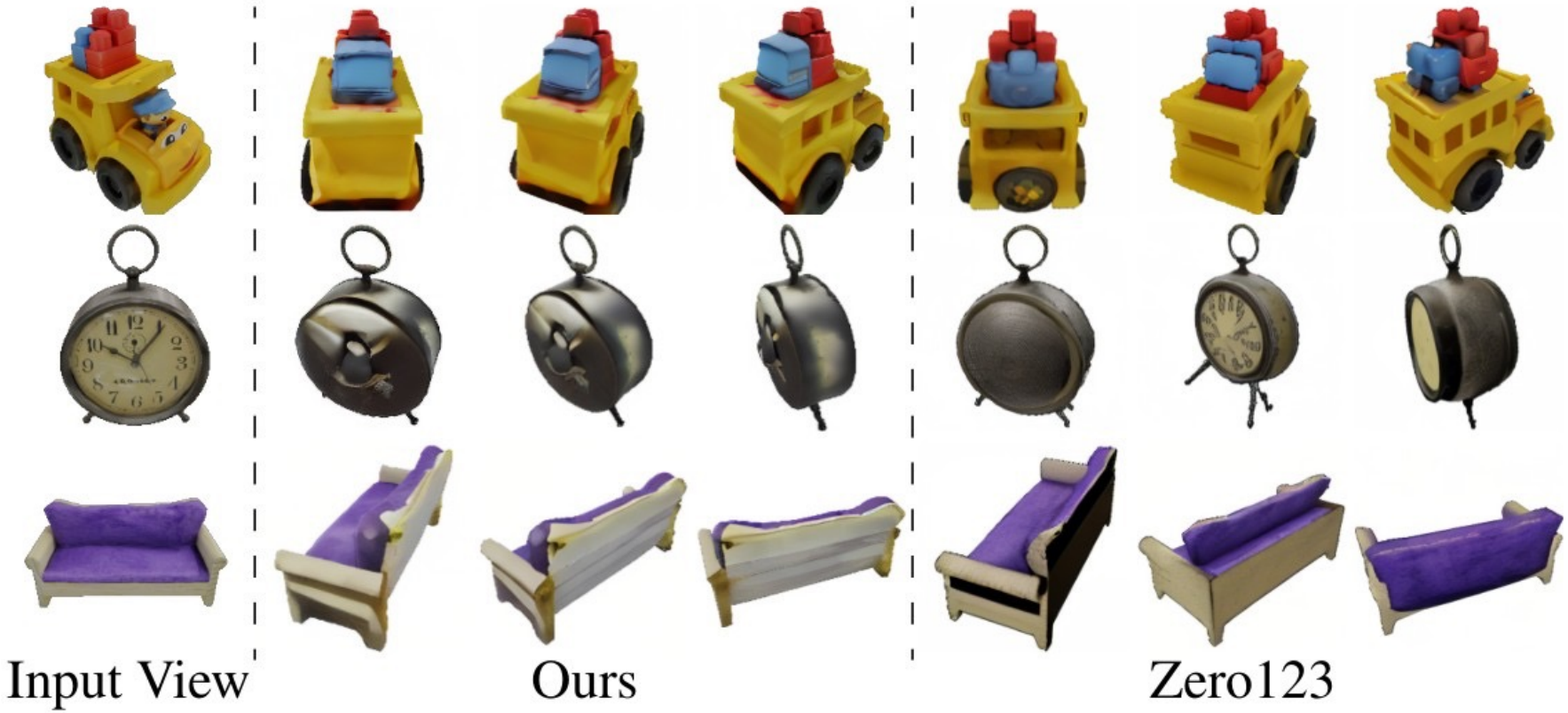
Liu, Ruoshi, et al. "Zero-1-to-3: Zero-shot one image to 3d object." *ICCV*. 2023.

Preliminary: Zero123

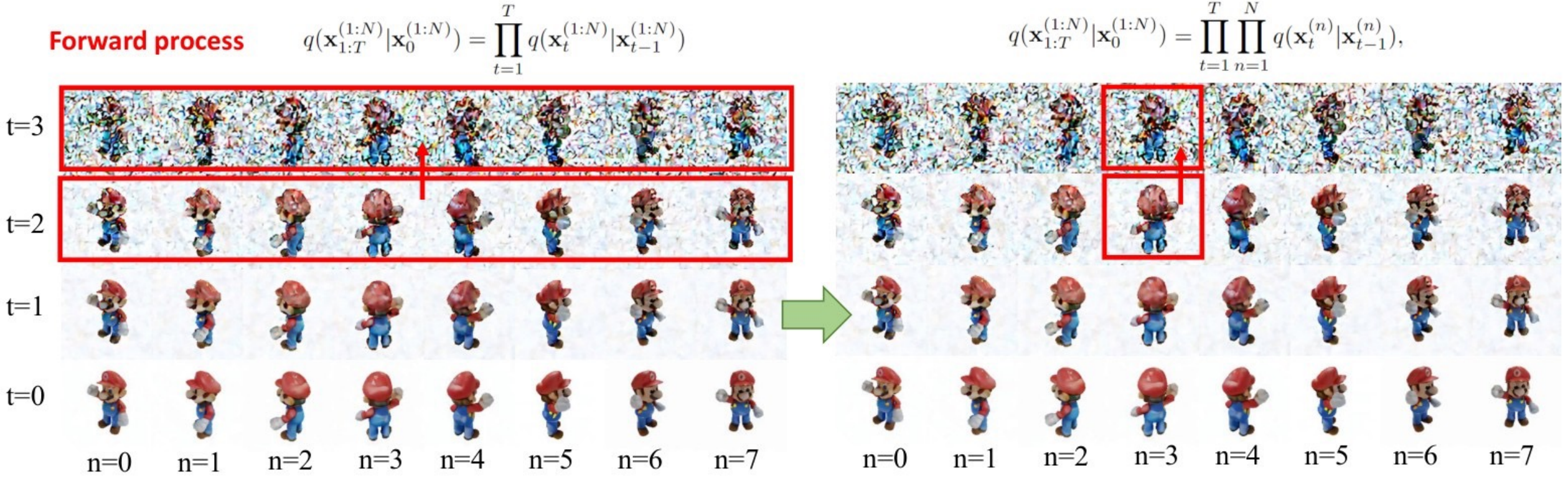


Liu, Ruoshi, et al. "Zero-1-to-3: Zero-shot one image to 3d object." *ICCV*. 2023.

Problem of Zero123: **Multiview Inconsistency**

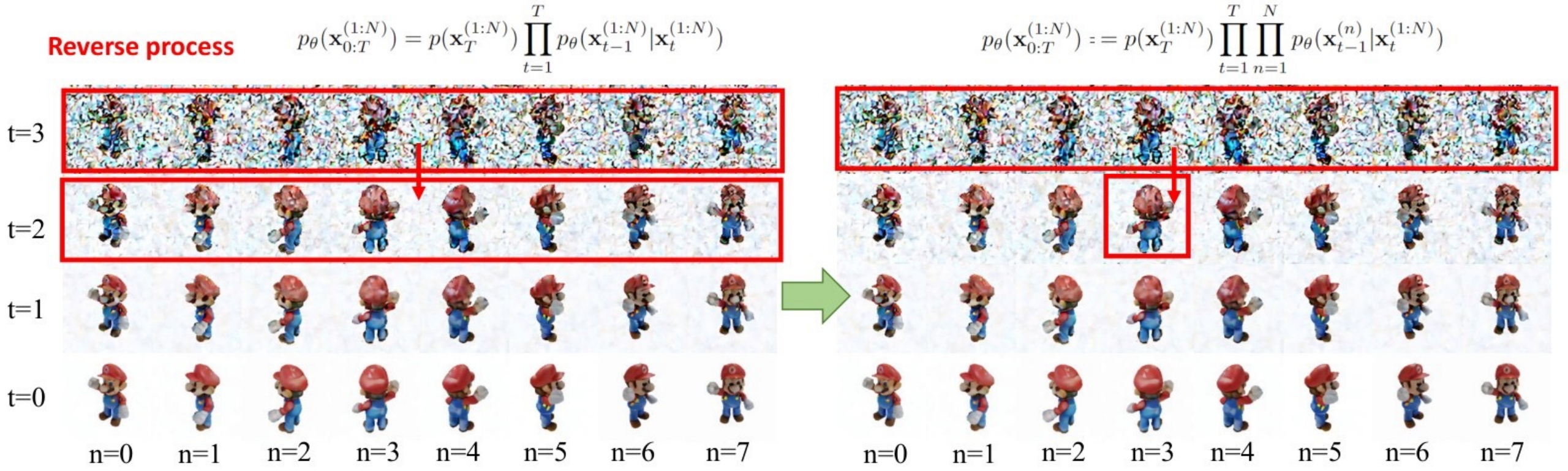


Method: Multiview Diffusion

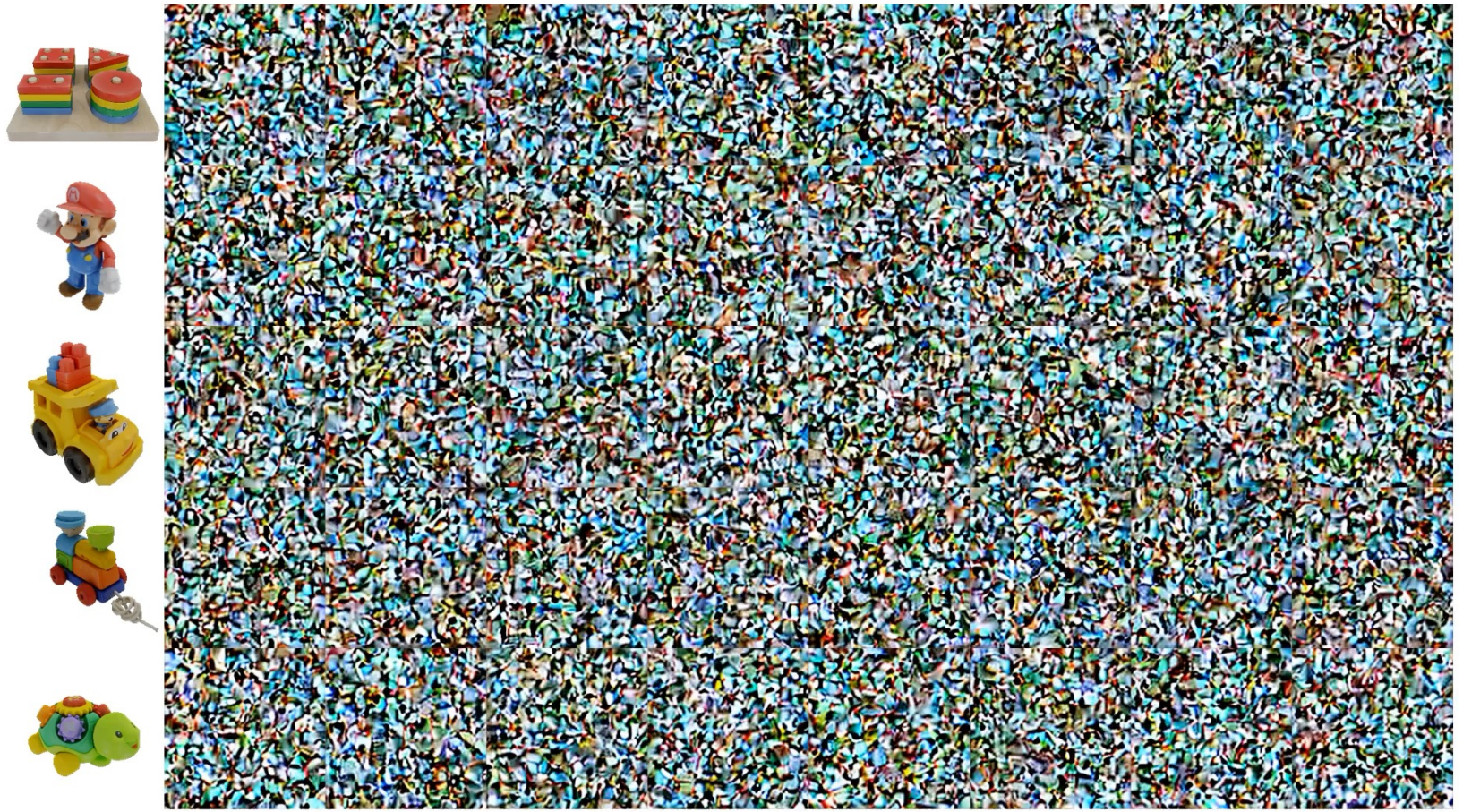


Method: Multiview Diffusion

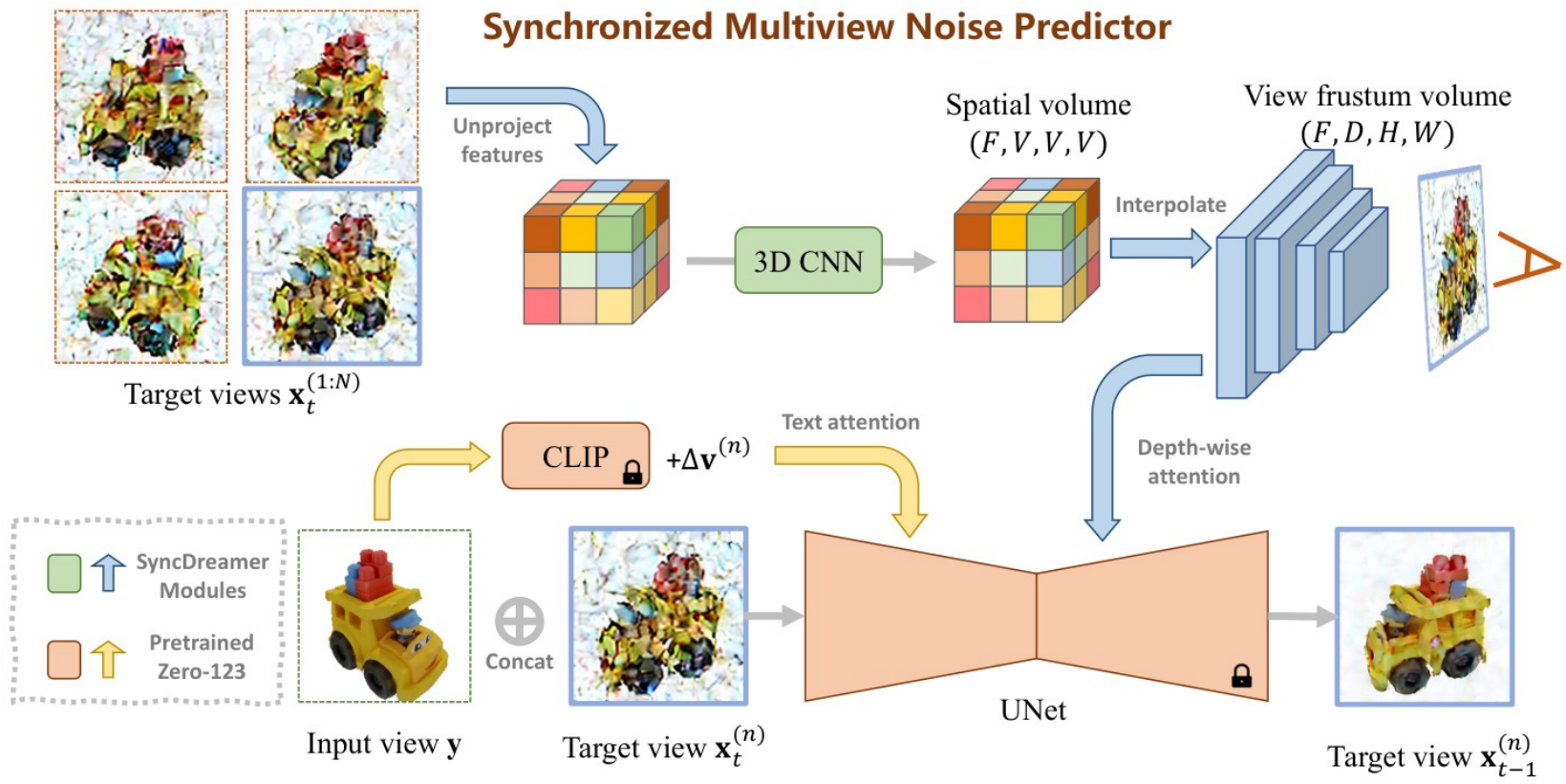
Synchronization: 每一個 denoising step 每個 view 都看得到其他 view



Reverse Process

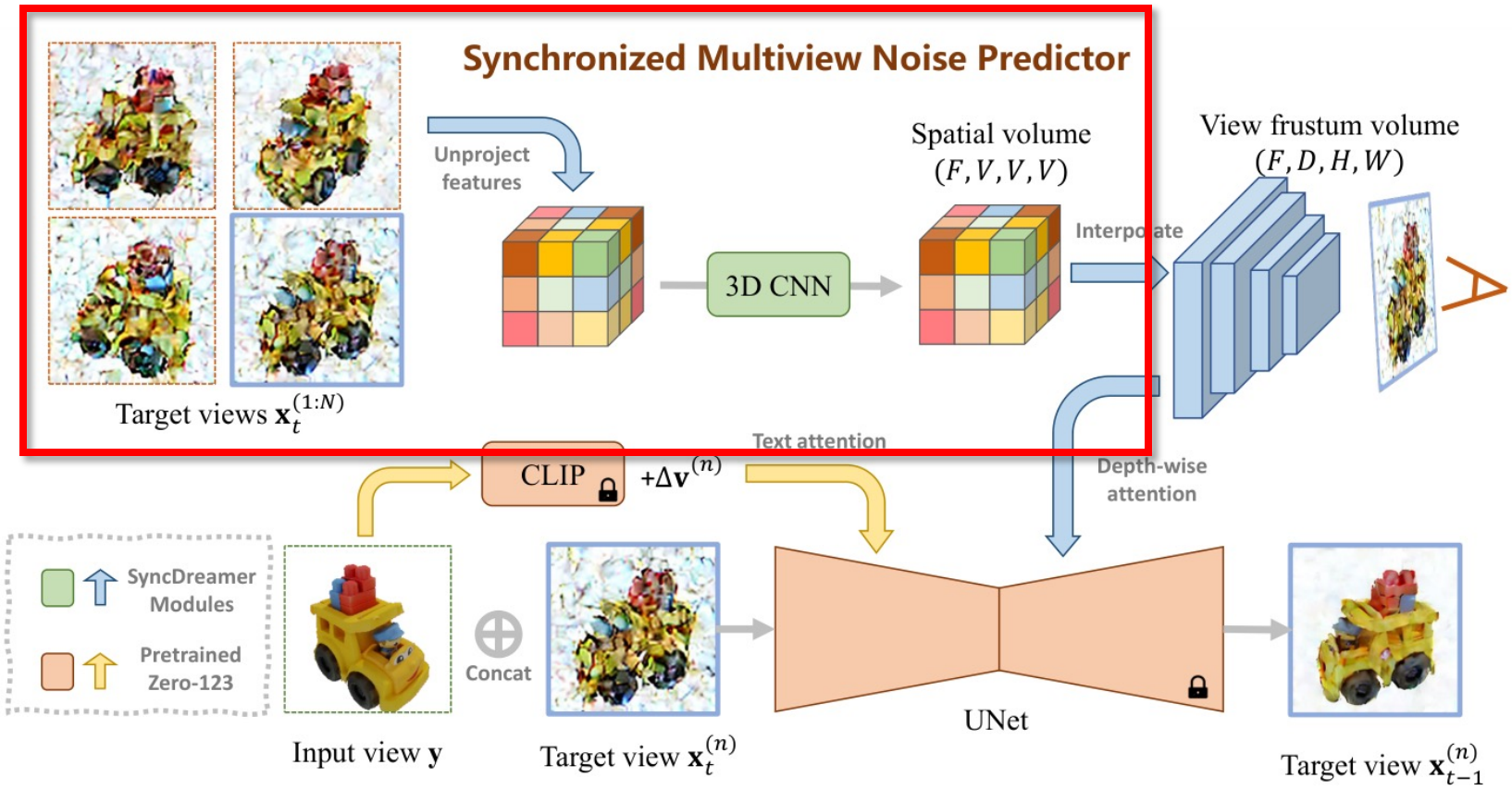


Method: 3D-aware Feature Attention



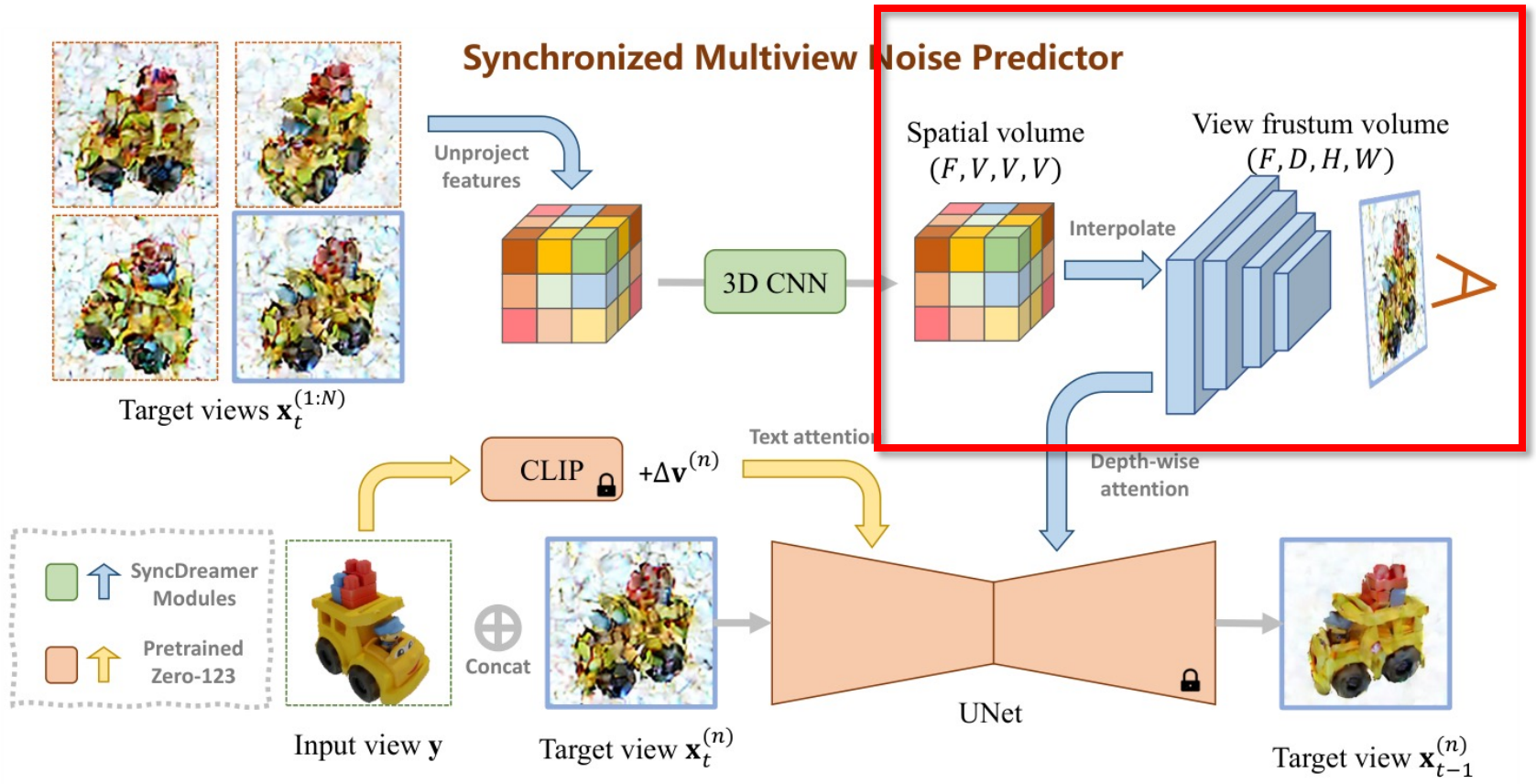
Method: 3D-aware Feature Attention

把 16 張 target views 的 noisy feature project 成 32x32x32 的 volume，用 3D CNN aggregate

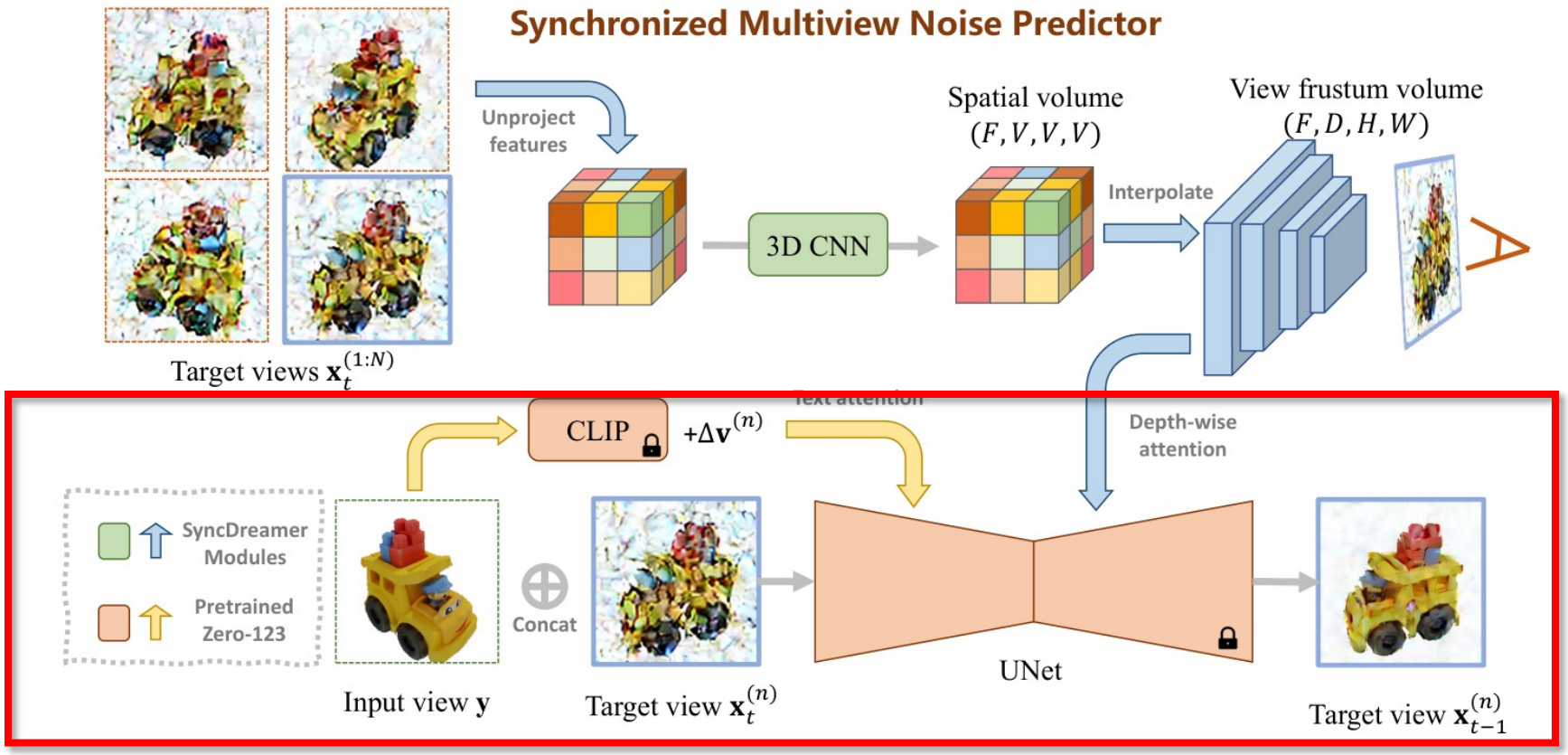


Method: 3D-aware Feature Attention

沿著這個 target view 的每個 pixel 取 48 個 depth，從 volume 內插出 32x32x48 的 volume



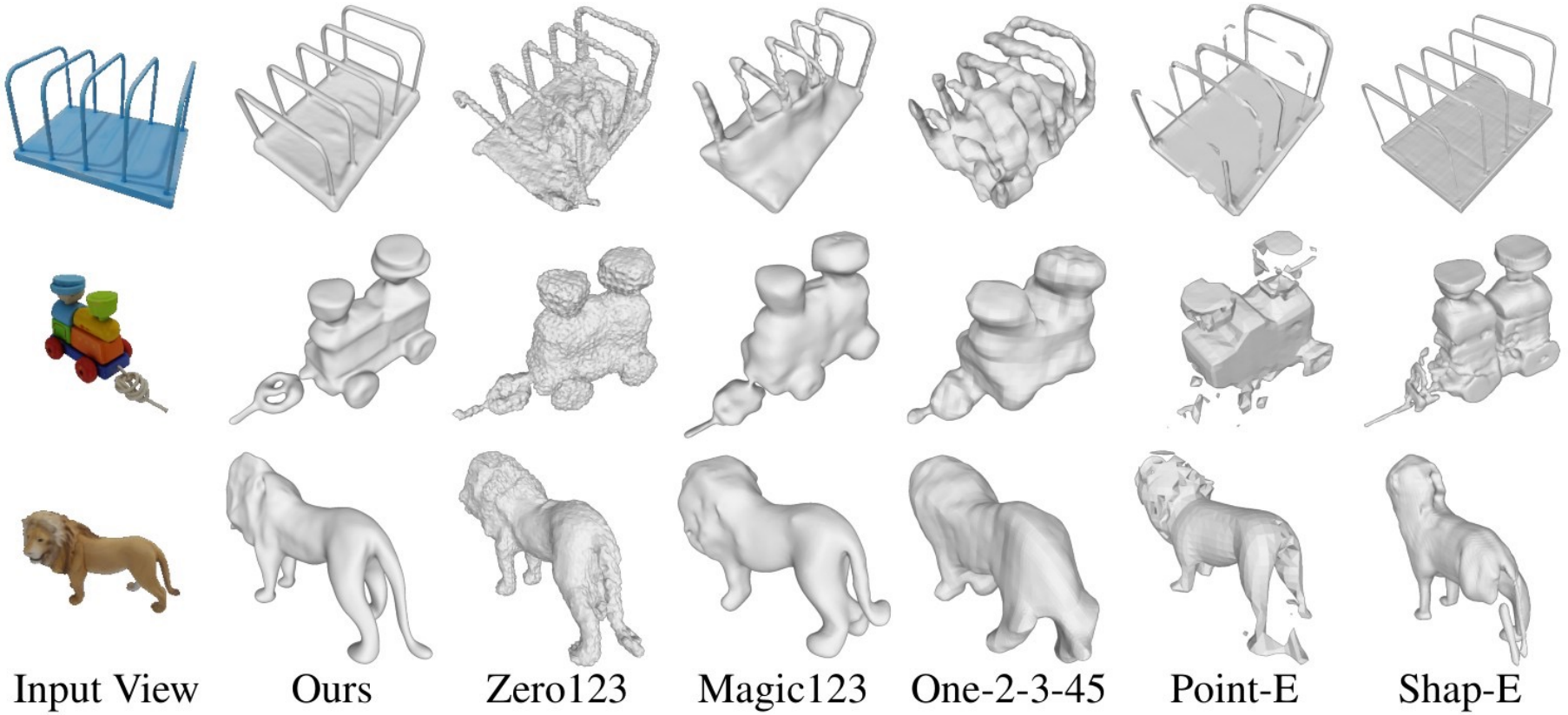
Method: 3D-aware Feature Attention



用 frozen 的 zero123 UNet 去 predict noise

上面得到的 view frustum volume 用 cross attention 來 inject

Experiments (NeuS)



More Results



Summary

從單張影像生成 3D 物體，這時候 correspondence 完全未知，無法應用 MultiDiffusion

改成 training-based 的做法，但沿用現有的 single image -> 3D object model Zero123，加入一個**所有 view 共享的 volume**（一個 joint distribution）來維持 consistency、也讓所有 view 互相交換資訊

LookingGlass

Chang, Pascal, et al. "LookingGlass: Generative Anamorphoses via Laplacian Pyramid Warping." *CVPR.* 2025.

Anamorphoses 變形畫



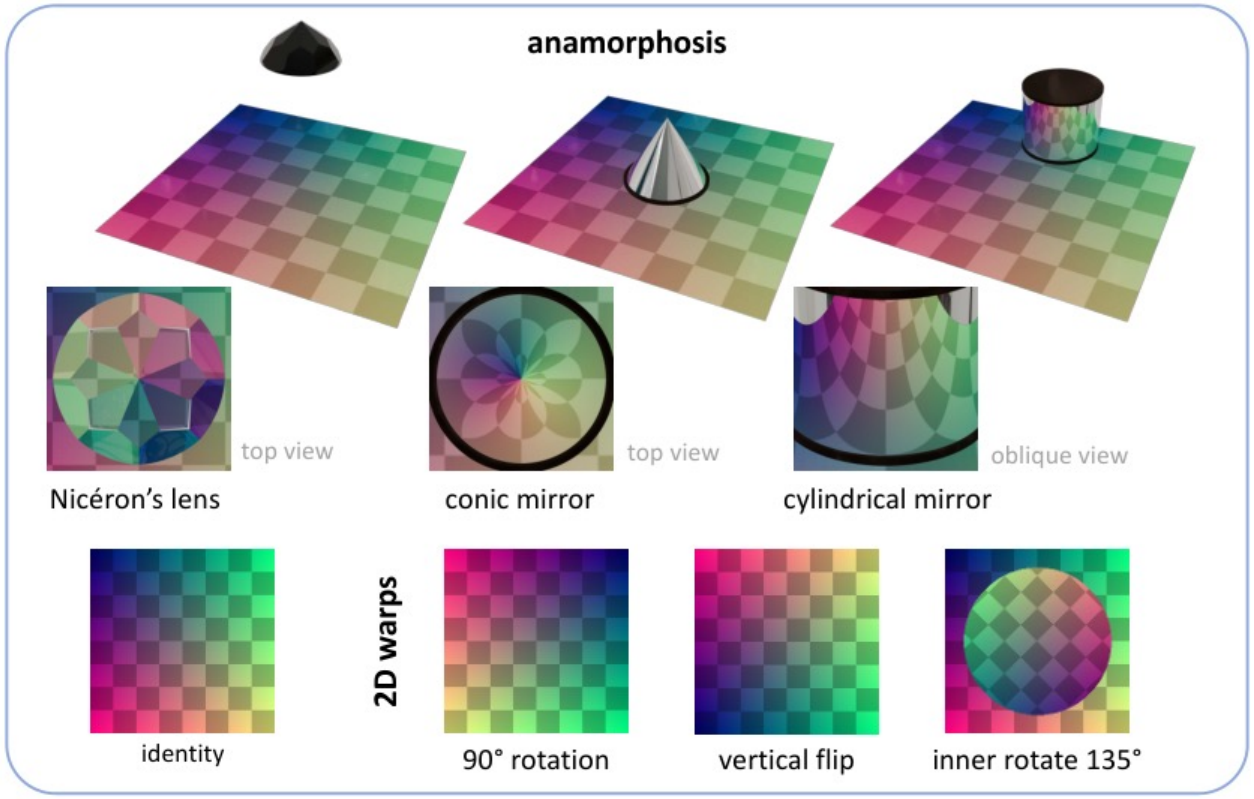
A Real Life Demo



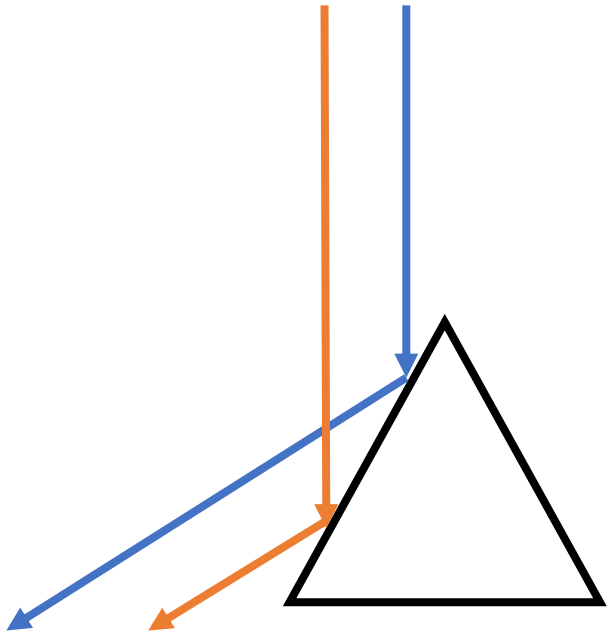




Challenge

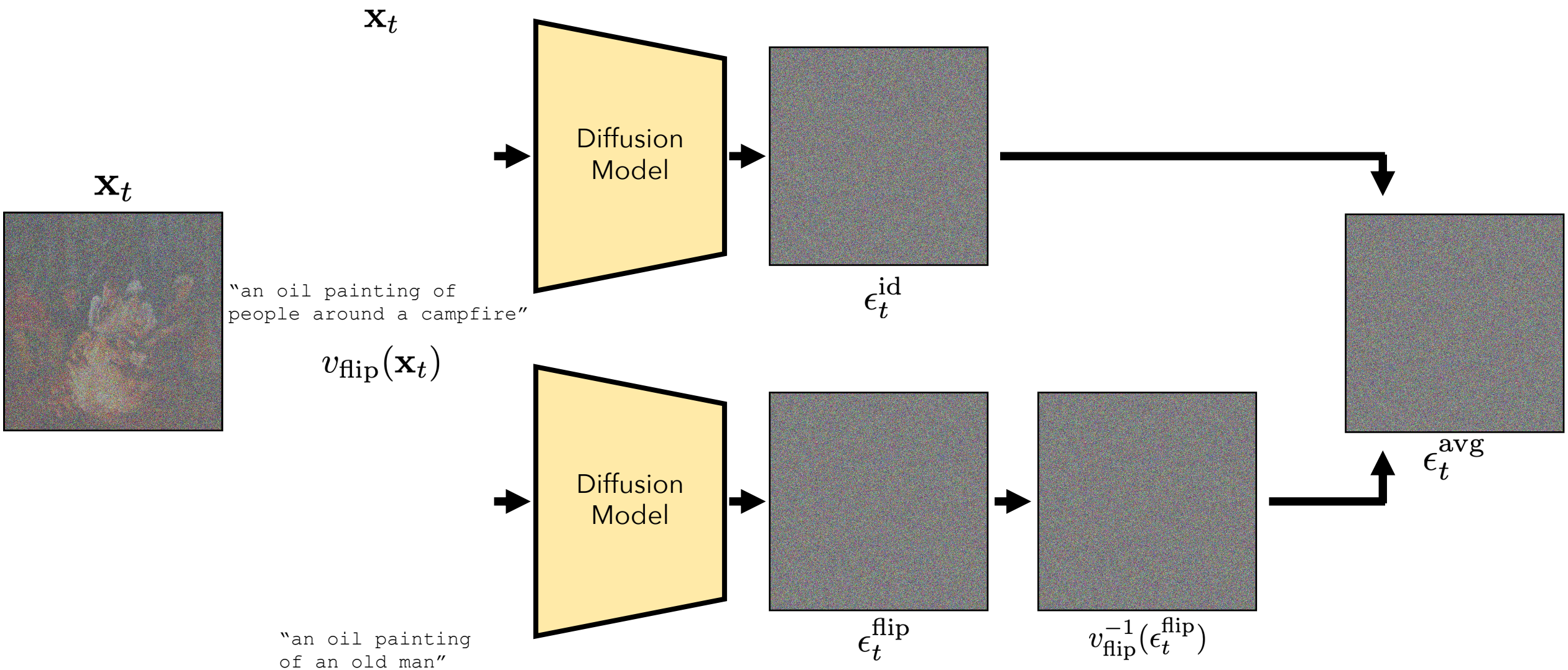


(c) Views considered

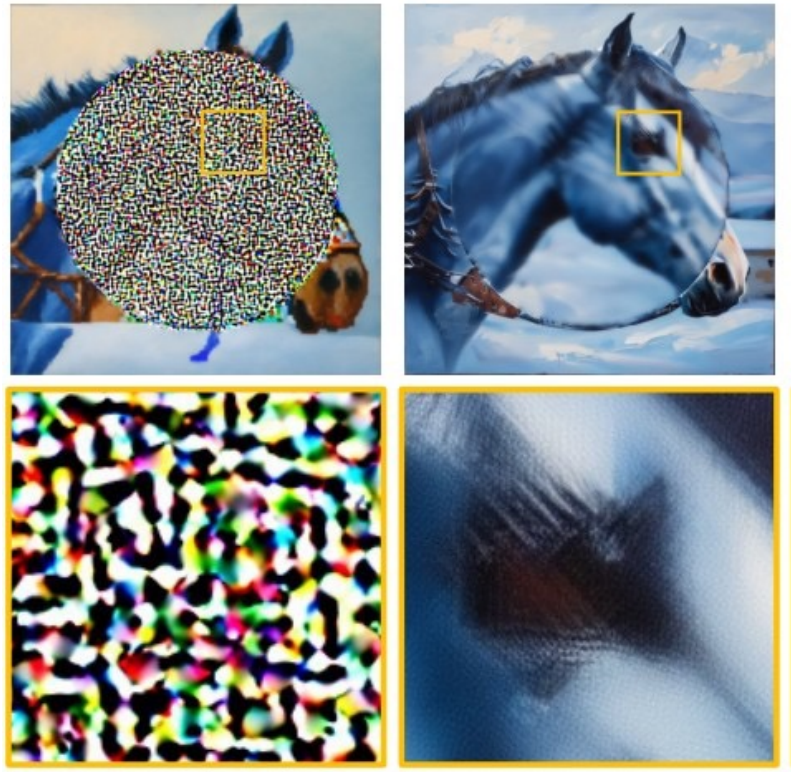


鏡中與原始畫布看到的內外是反過來的

Recap: Visual Anagrams

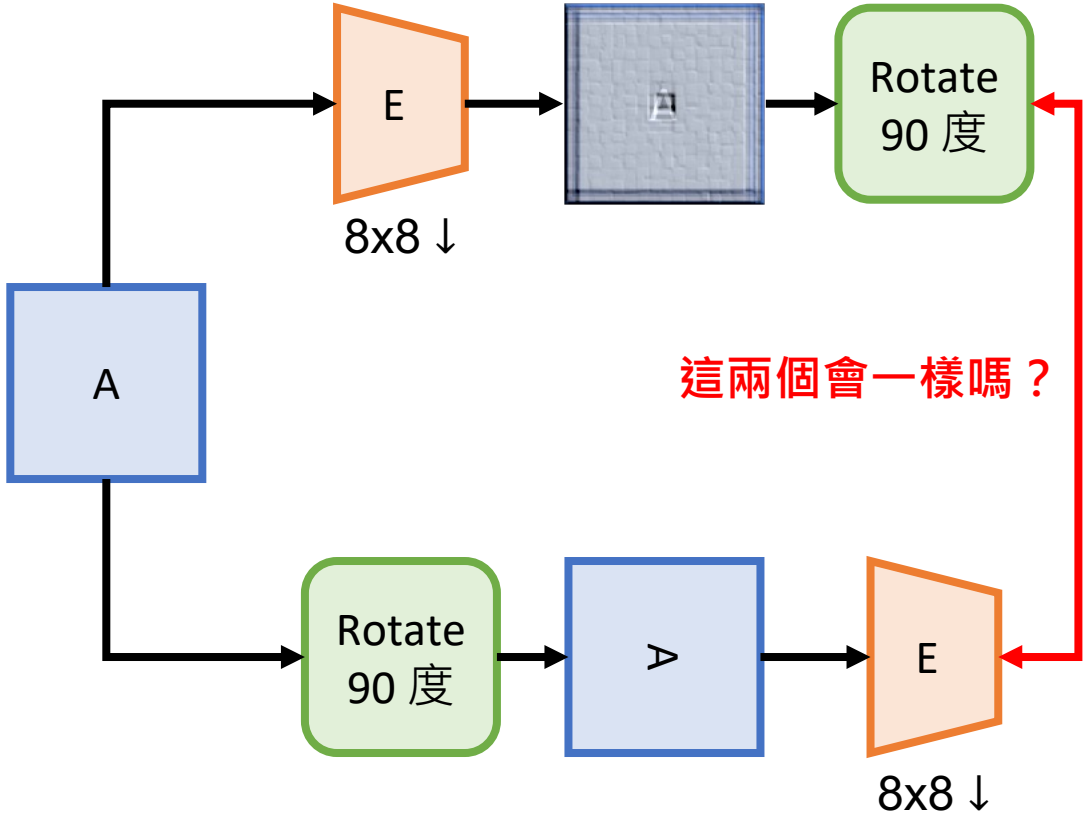


Problems: Latent Visual Anagrams



Geng *et al.*

+ Final Estimate



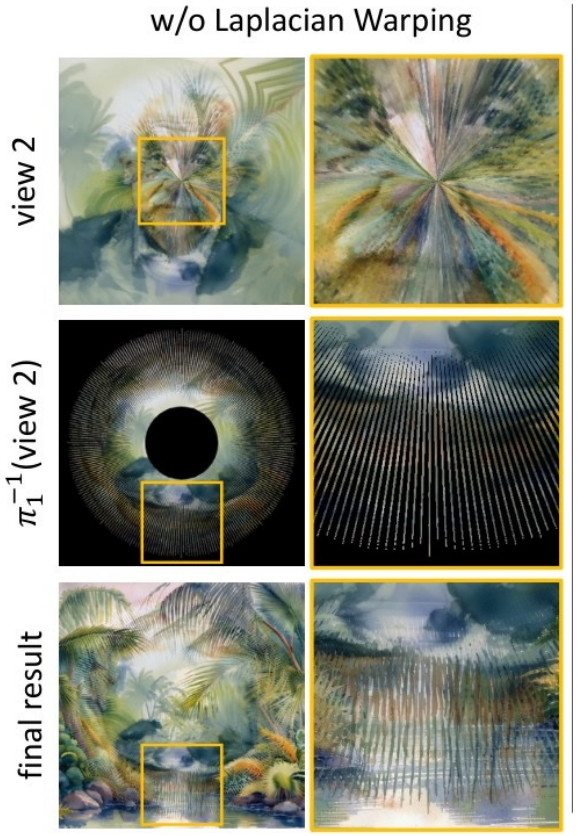
因為 VAE 對於 warping (也包含各種旋轉) 並不 equivariant

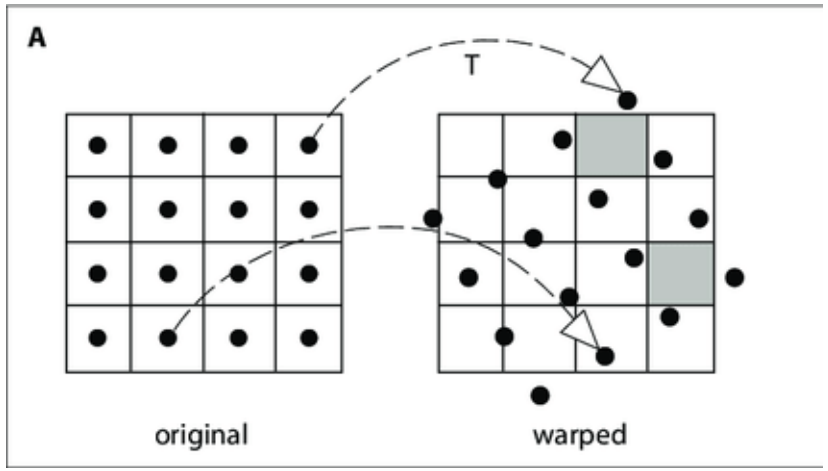
-> 無法像 Visual Anagrams 那樣直接在 latent space 用 MultiDiffusion 平均大法

Problems: Warping

鏡中的圖

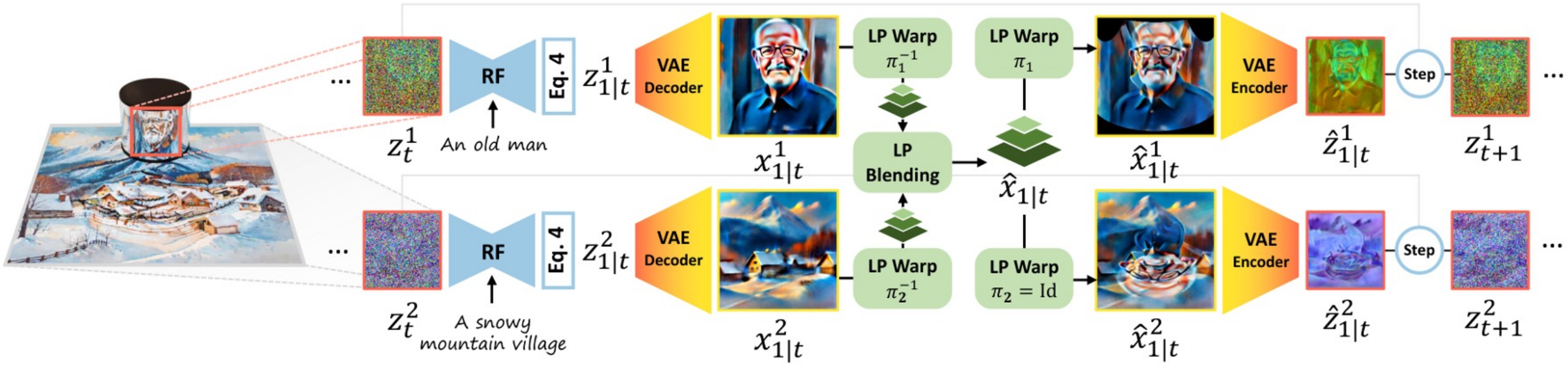
Warp 到原始畫布上



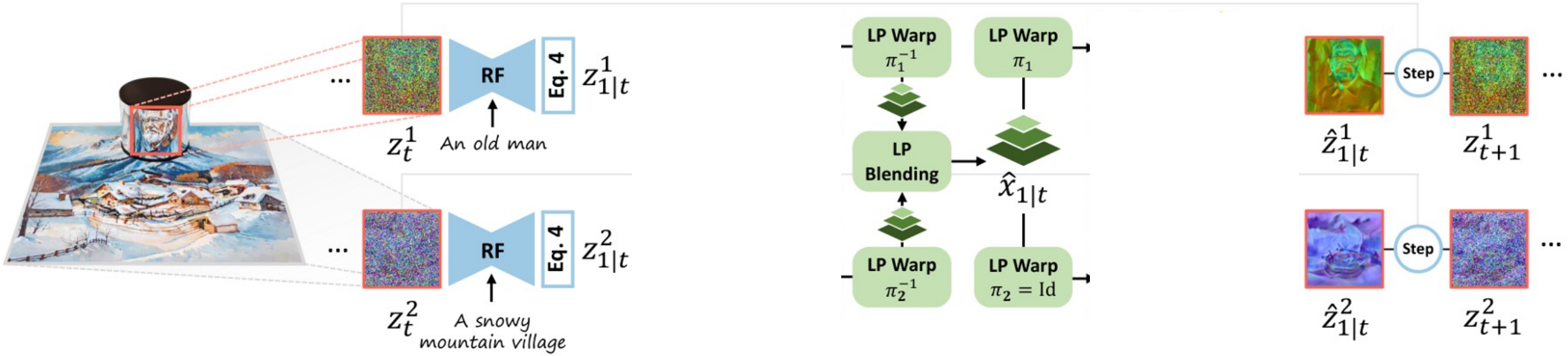


Forward warping 會造成有些位置有空洞

Method


RF
Eq. 4
An old man
A snowy mountain village
VAE Decoder
LP Warp π_1^{-1}
LP Warp π_2^{-1}
LP Blending
LP Warp π_1
LP Warp $\pi_2 = \mathrm{Id}$
VAE Encoder
Step
z_t^1
z_t^2
$z_{1|t}^1$
$z_{1|t}^2$
$x_{1|t}^1$
$x_{1|t}^2$
$\hat{x}_{1|t}$
$\hat{x}_{1|t}^1$
$\hat{x}_{1|t}^2$
$\hat{z}_{1|t}^1$
$\hat{z}_{1|t}^2$
z_{t+1}^1
z_{t+1}^2

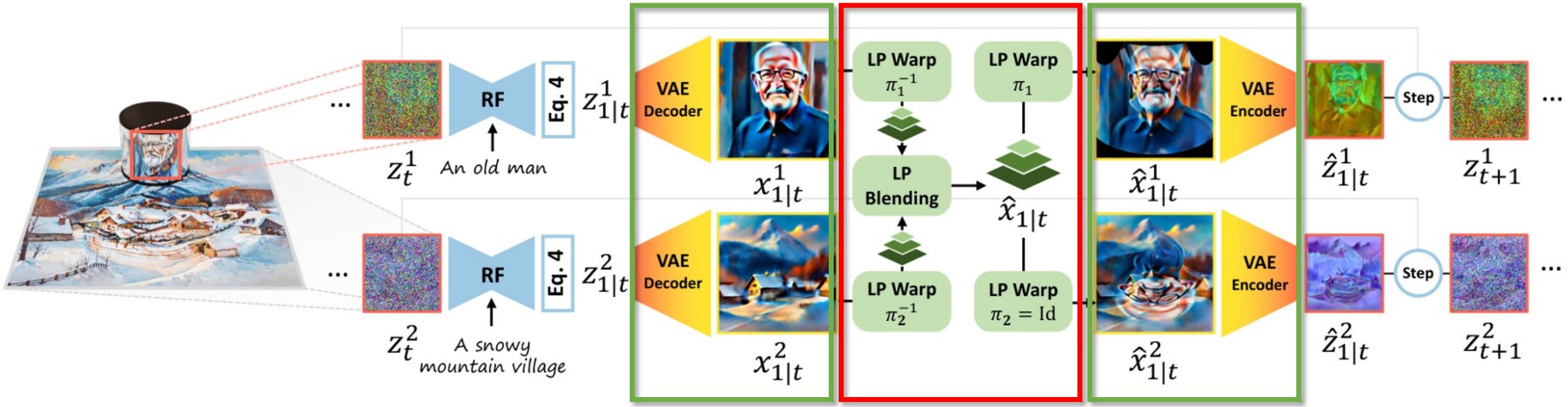
Method



這樣就長得跟 Visual Anagrams 很像

Method

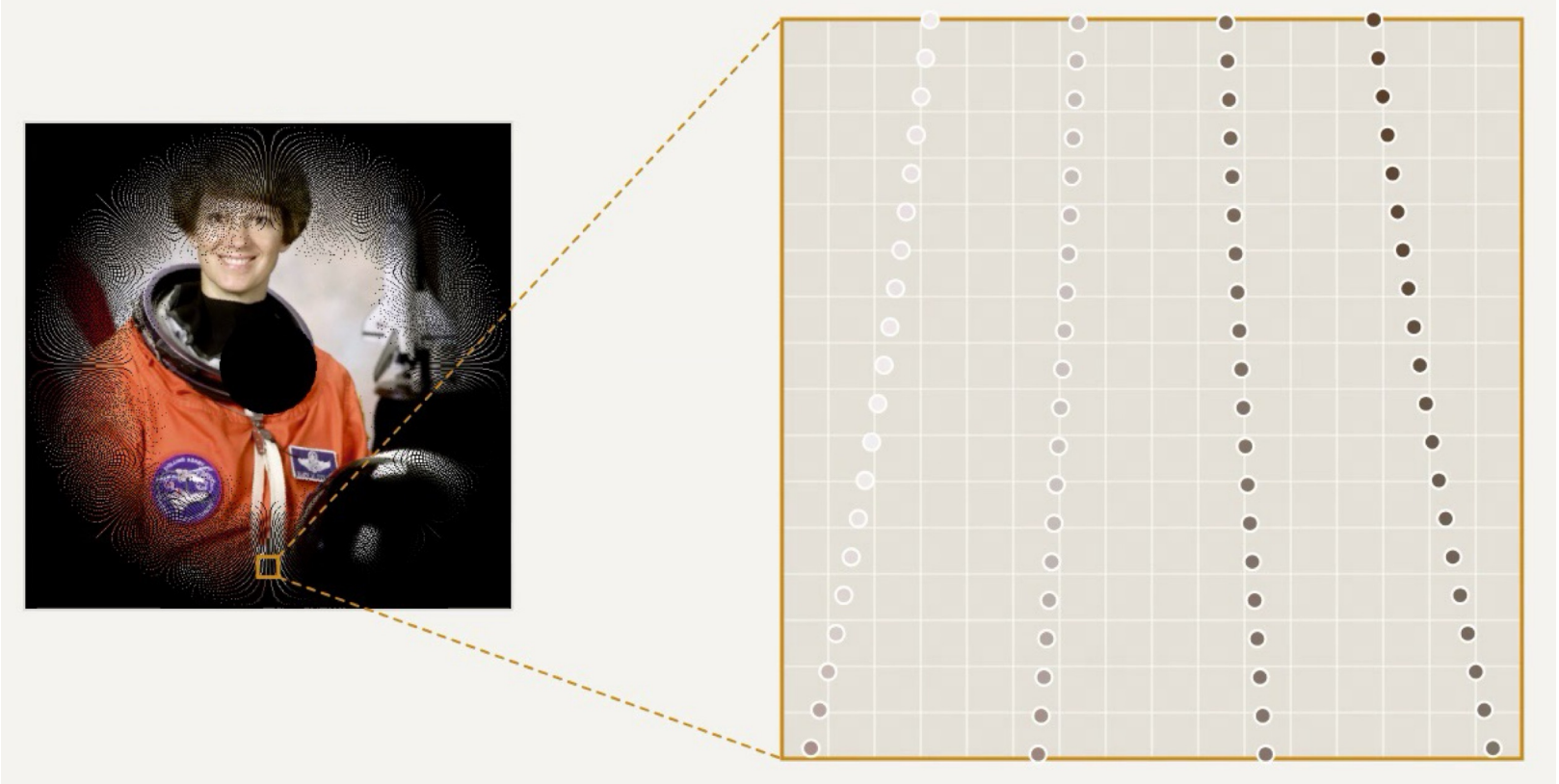
所有跟 warping 與平均有關的操作都必須在 pixel space 做



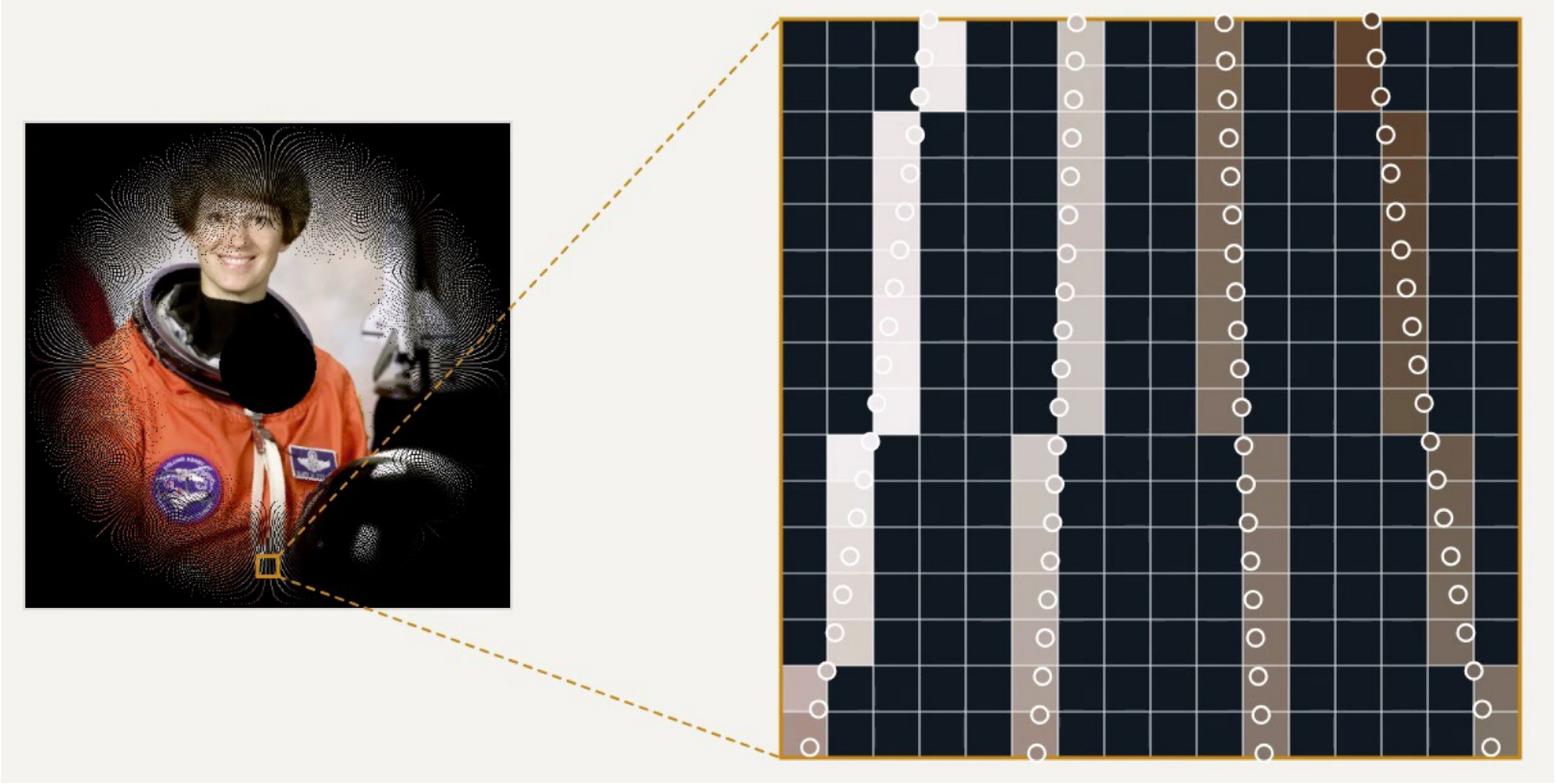
Latent space 不能做 warping/averaging，但可以用 encoder/decoder 轉換到 pixel space

MeDM 其實也用了一樣的設計，而且更早（但沒有被引用 QQ）

Method: Laplacian Pyramid Warping

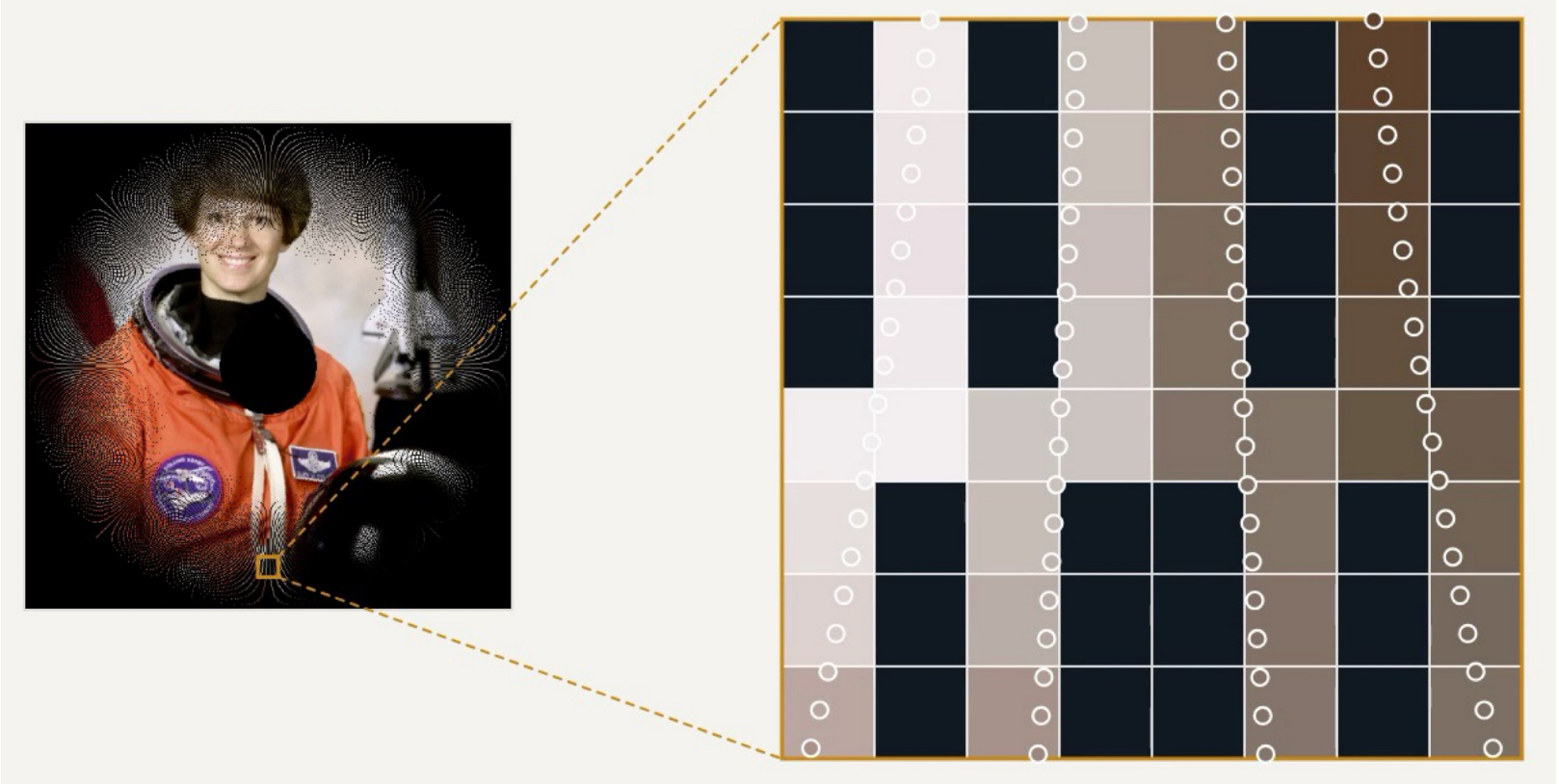


Method: Laplacian Pyramid Warping



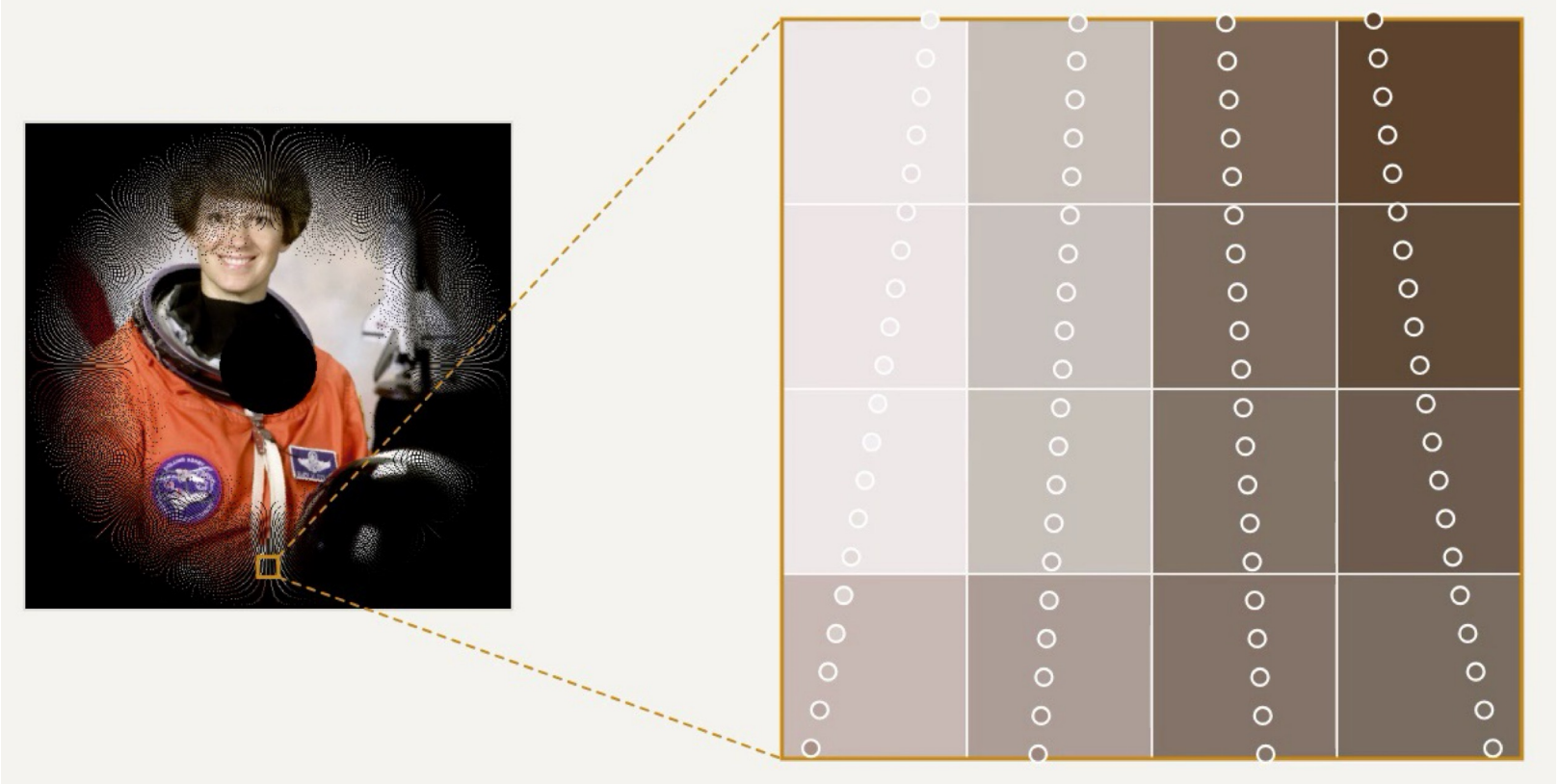
因為 sample 之間的 gap 比 pixel 的間隔更大，所以會有 pixel 位置沒有任何 sample 落入，就產生黑洞

Method: Laplacian Pyramid Warping



把 pixel 的大小放大一倍（也就是用較低的解析度），就可以把一些洞補起來

Method: Laplacian Pyramid Warping

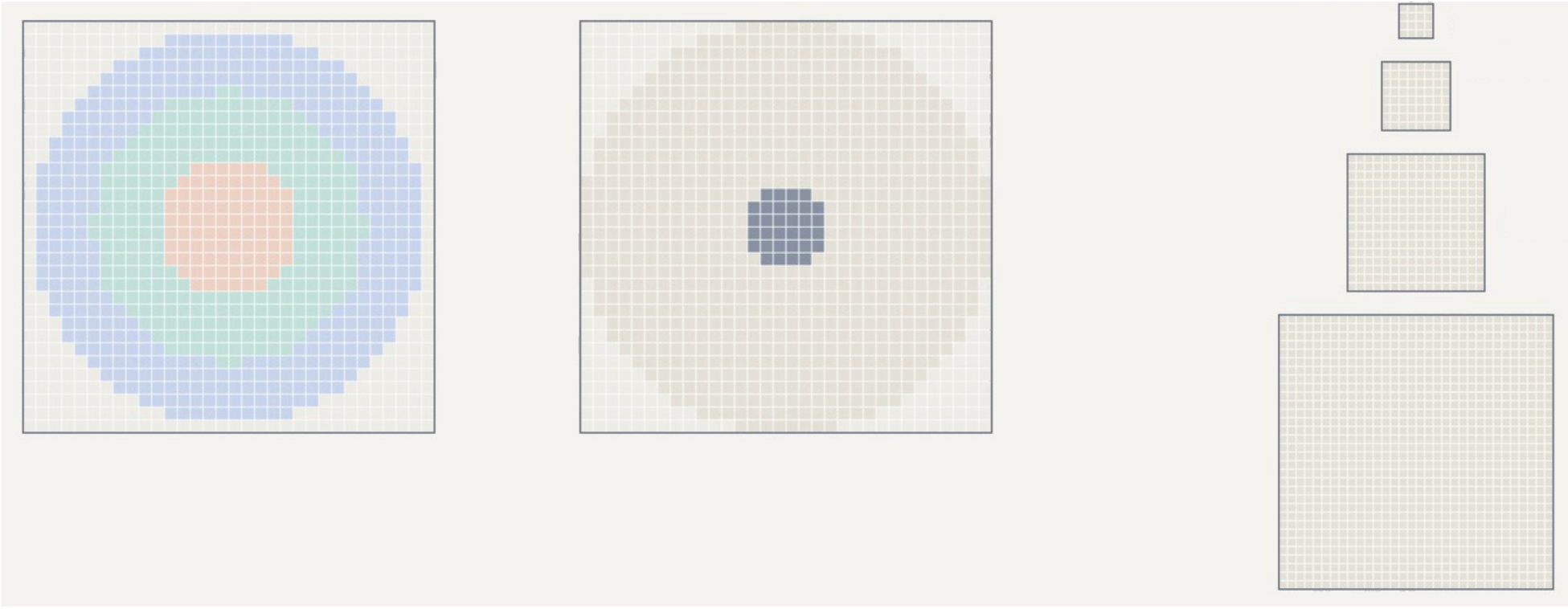


用更低的解析度，所有的洞都被補起來

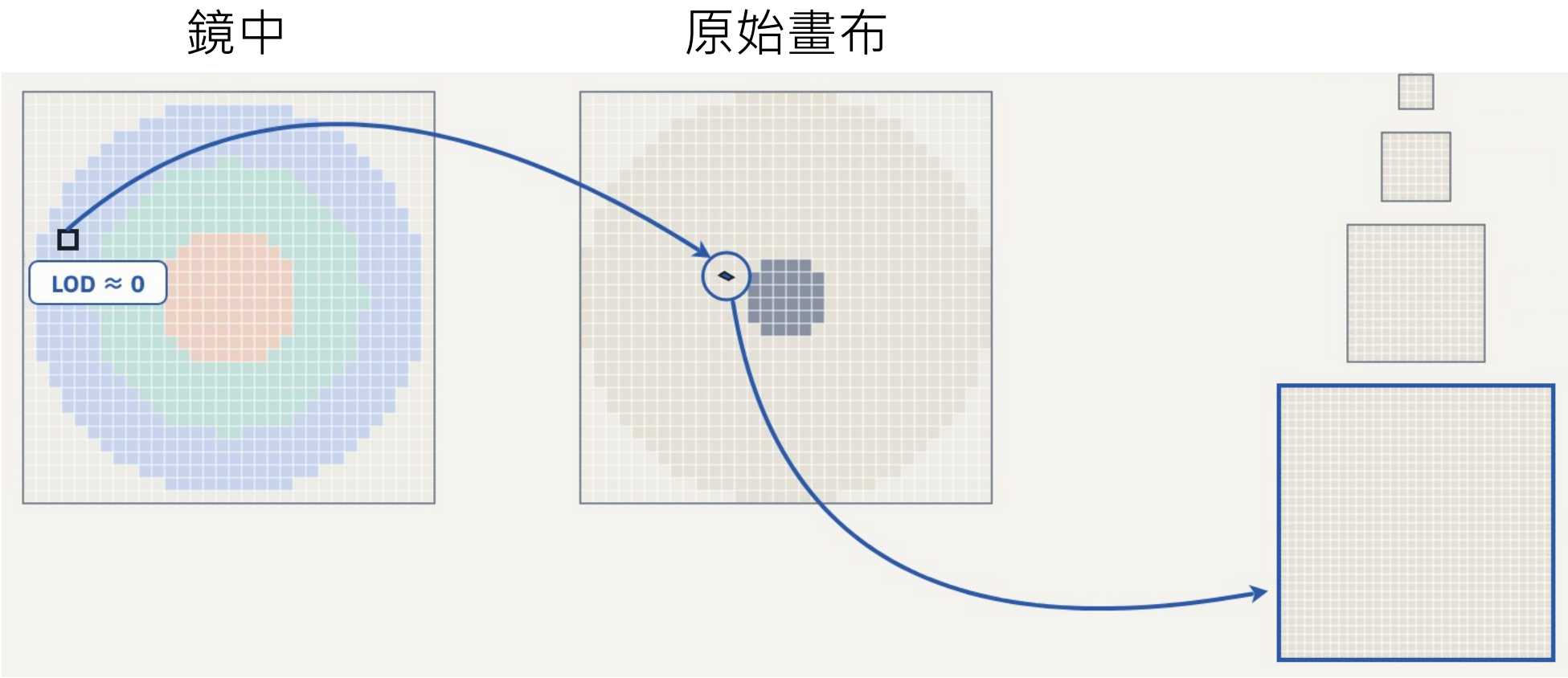
Method: Level of Detail (LOD)

鏡中

原始畫布

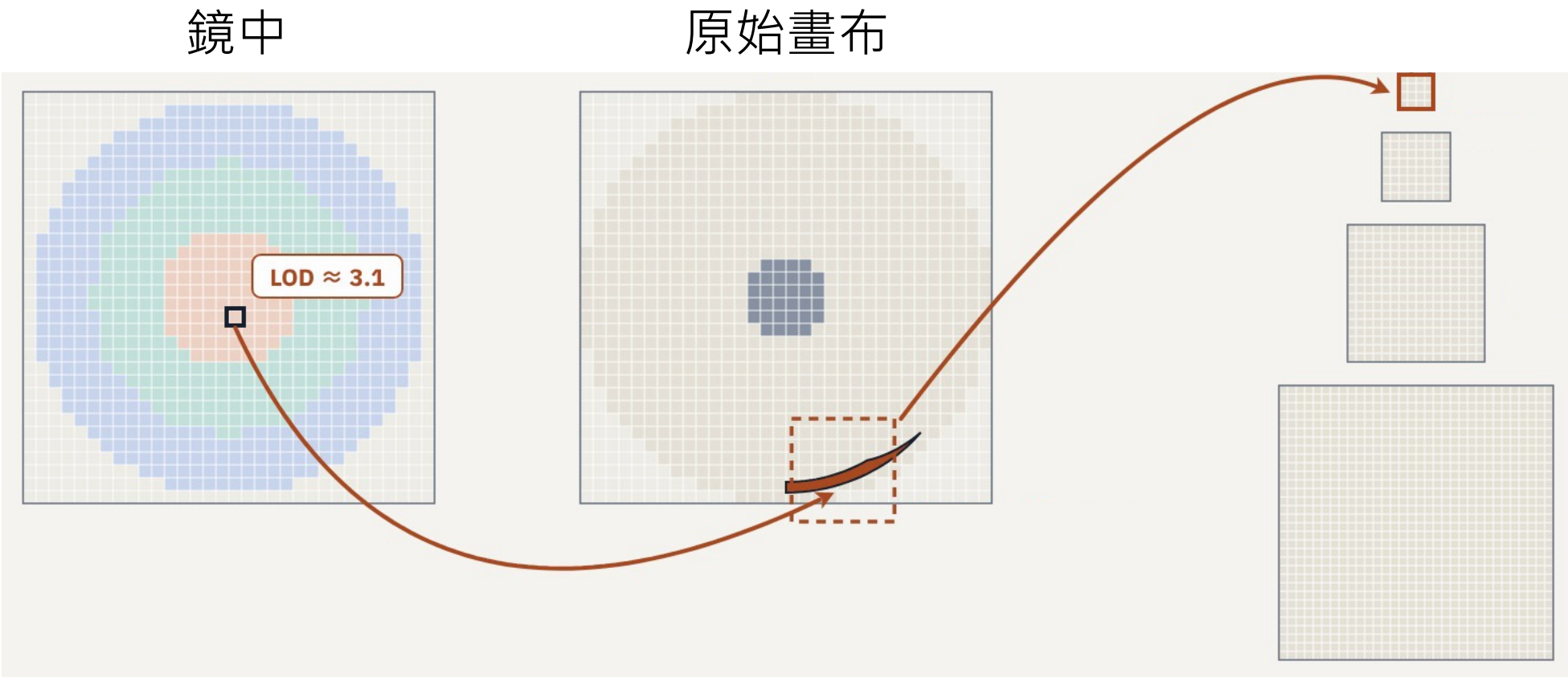


Method: Level of Detail (LOD)



靠近鏡中外圈，對應到原始畫布會**集中**，所以可用比較**高解析度**（精細）的 map

Method: Level of Detail (LOD)

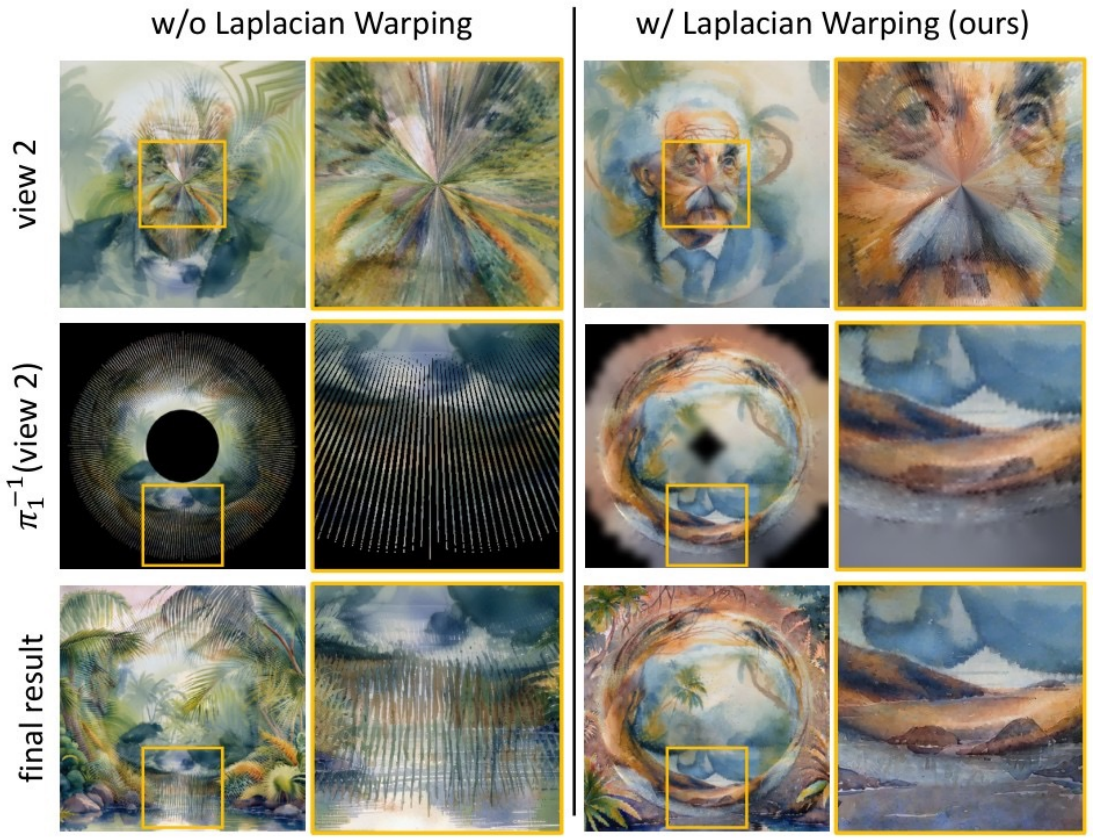


靠近鏡中中心，對應到原始畫布會**散開**，所以要用比較**低解析度**（粗糙）的 map

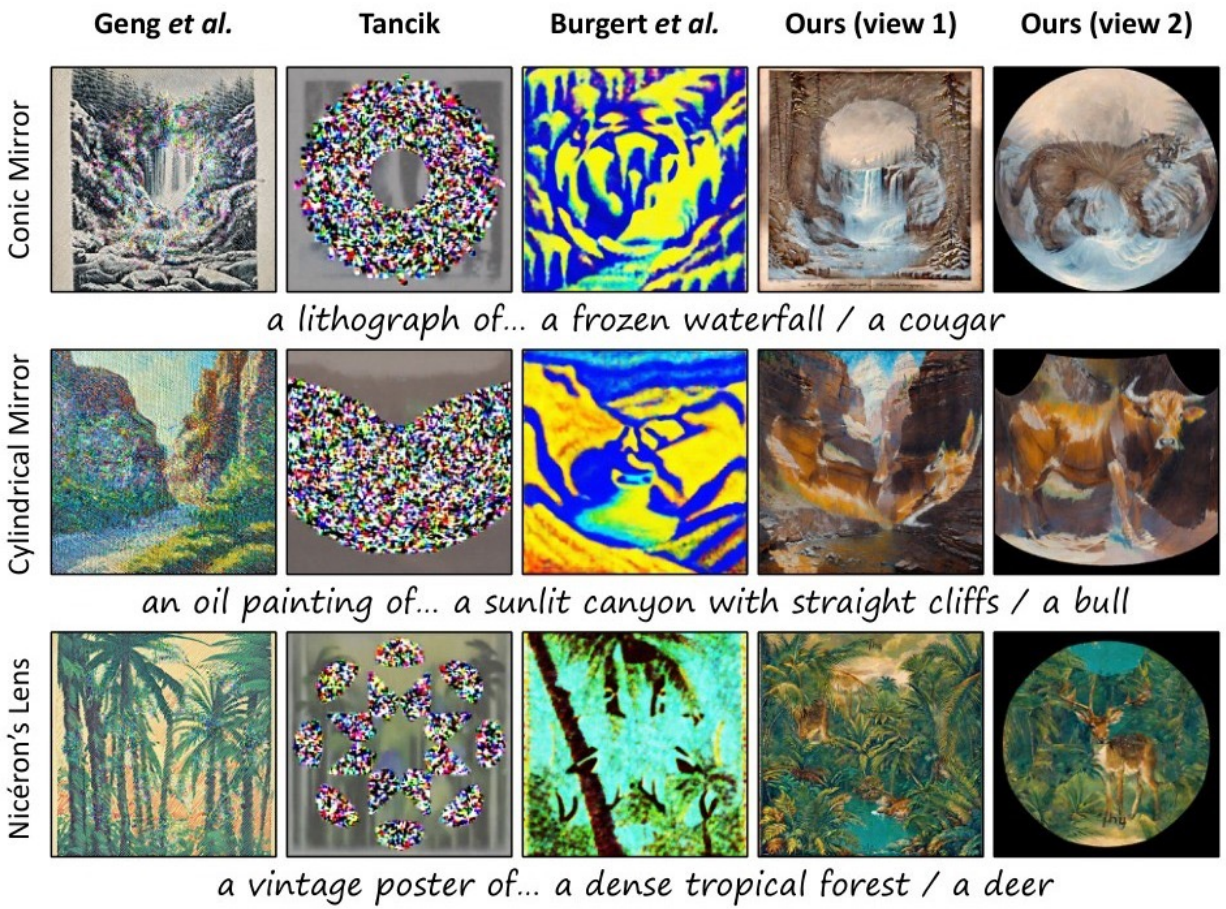
Method: Laplacian Pyramid Warping

鏡中的圖

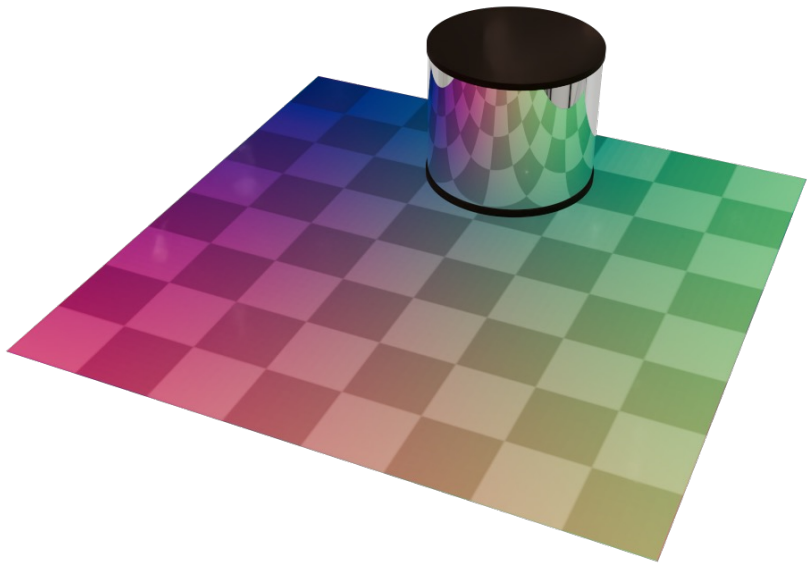
Warp 到原始畫布上



Experiments



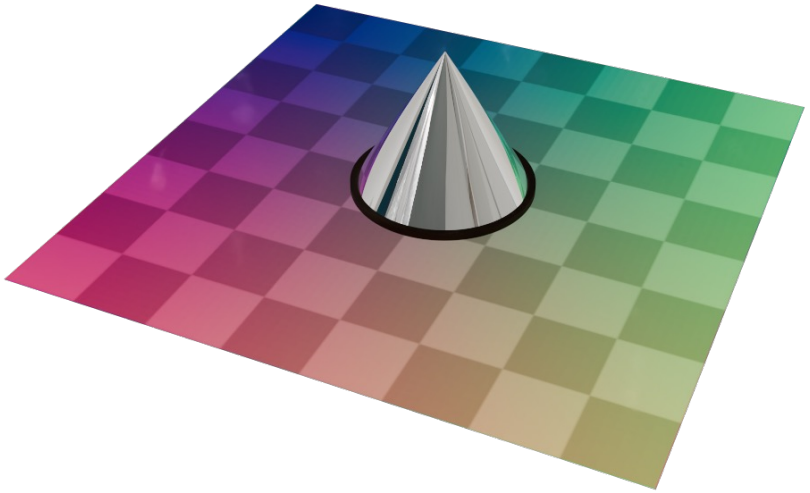
More Results: Cylindrical mirror





an oil painting of a sunlit canyon with straight cliffs

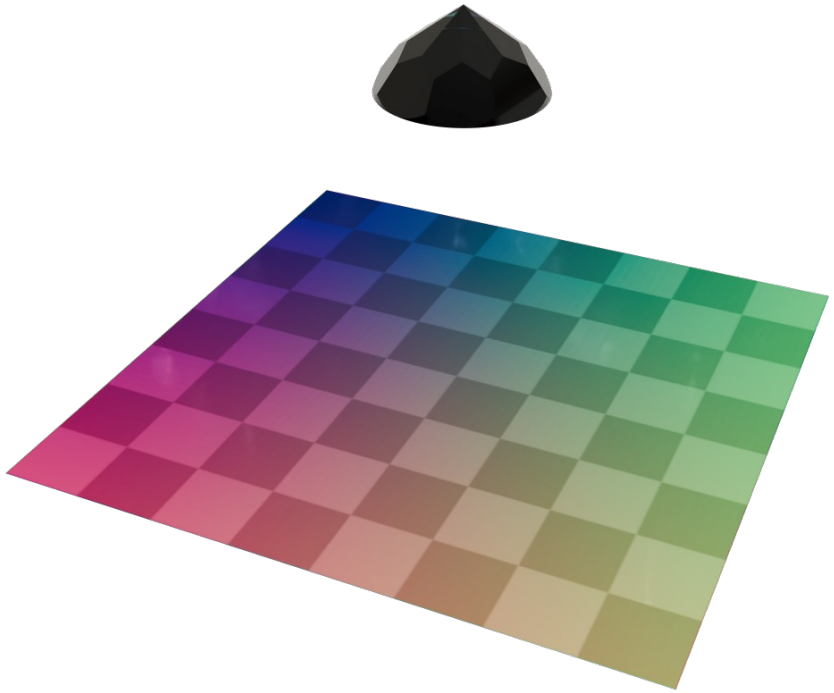
More Results: Conic mirror





a pixel art version of flower petals close-up

More Results: Nicéron's lens





a lithograph of a horizon of a wheat field

Summary

當 **correspondence 不是 1-to-1 mapping** 時，要先把每個 view 轉成 Laplacian pyramid，再依照局部的壓縮/拉伸程度從不同的 pyramid 層 sample 資訊